CHAPTER

7

EXTERNAL FORCED CONVECTION

In Chap. 6, we considered the general and theoretical aspects of forced convection, with emphasis on differential formulation and analytical solutions. In this chapter, we consider the practical aspects of forced convection to or from flat or curved surfaces subjected to *external flow,* characterized by the freely growing boundary layers surrounded by a free flow region that involves no velocity and temperature gradients.

We start this chapter with an overview of external flow, with emphasis on friction and pressure drag, flow separation, and the evaluation of average drag and convection coefficients. We continue with *parallel flow over flat plates.* In Chap. 6, we solved the boundary layer equations for steady, laminar, parallel flow over a flat plate, and we obtained relations for the local friction coefficient and the Nusselt number. Using these relations as the starting point, we determine the average friction coefficient and Nusselt number. We then extend the analysis to turbulent flow over flat plates with and without an unheated starting length.

Next, we consider *crossflow over cylinders and spheres,* and we present graphs and empirical correlations for the drag coefficients and the Nusselt numbers and discuss their significance. Finally, we consider *crossflow over tube banks* in aligned and staggered configurations, and we present correlations for the pressure drop and the average Nusselt number for both configurations.

OBJECTIVES

When you finish studying this chapter, you should be able to:

- Distinguish between internal and external flow,
- Develop an intuitive understanding of friction drag and pressure drag, and evaluate the average drag and convection coefficients in external flow,
- Evaluate the drag and heat transfer associated with flow over a flat plate for both laminar and turbulent flow,
- Calculate the drag force exerted on cylinders and spheres during crossflow, and the average heat transfer coefficient, and
- Determine the pressure drop and the average heat transfer coefficient associated with flow across a tube bank for both in-line and staggered configurations.

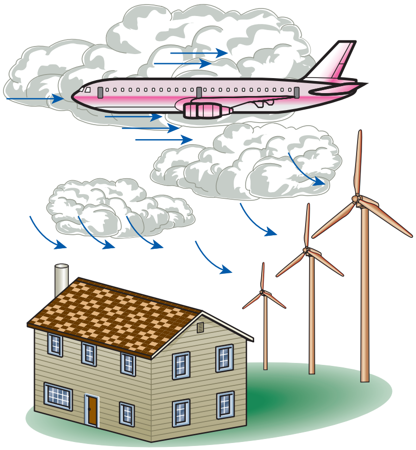

FIGURE 7–1
Flow over bodies is commonly encountered in practice.

7–1 ■ DRAG AND HEAT TRANSFER IN EXTERNAL FLOW

Fluid flow over solid bodies often occurs in practice, and it is responsible for many physical phenomena such as the *drag force* acting on automobiles, power lines, trees, and underwater pipelines; the *lift* developed by airplane wings; the *upward draft* of rain, snow, hail, and dust particles in high winds; and the *cooling* of metal or plastic sheets, steam and hot water pipes, and extruded wires (Fig. 7–1). Therefore, developing a good understanding of external flow and external forced convection is important in the mechanical and thermal design of many engineering systems such as aircraft, automobiles, buildings, electronic components, and turbine blades.

The flow fields and geometries for most external flow problems are too complicated to be solved analytically, and thus we have to rely on correlations based on experimental data. The availability of high-speed computers has made it possible to conduct series of "numerical experimentations" quickly by solving the governing equations numerically and then to resort to expensive and time-consuming testing and experimentation only in the final stages of design. In this chapter, we mostly rely on relations developed experimentally.

The velocity of the fluid relative to an immersed solid body sufficiently far from the body (outside the boundary layer) is called the **free-stream velocity**. It is usually taken to be equal to the **upstream velocity** V, also called the **approach velocity**, which is the velocity of the approaching fluid far ahead of the body. This idealization is nearly exact for very thin bodies, such as a flat plate parallel to flow, but approximate for blunt bodies such as a large cylinder. The fluid velocity ranges from zero at the surface (the no-slip condition) to the free-stream value away from the surface, and the subscript "infinity" serves as a reminder that this is the value at a distance where the presence of the body is not felt. The upstream velocity, in general, may vary with location and time (e.g., the wind blowing past a building). But in the design and analysis, the upstream velocity is usually assumed to be *uniform* and *steady* for convenience, and this is what we will do in this chapter.

Friction and Pressure Drag

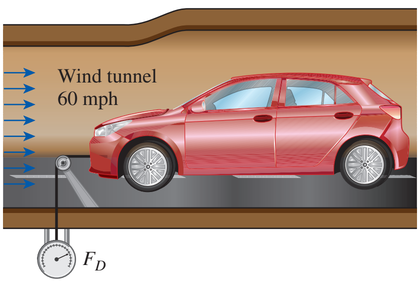

FIGURE 7–2
Schematic for measuring the drag force acting on a car in a wind tunnel.

It is common experience that a body meets some resistance when it is forced to move through a fluid, especially a liquid. You may have seen high winds knocking down trees, power lines, and even trailers, and have felt the strong "push" the wind exerts on your body. You experience the same feeling when you extend your arm out of the window of a moving car. The force a flowing fluid exerts on a body in the flow direction is called **drag** (Fig. 7–2).

A stationary fluid exerts only normal pressure forces on the surface of a body immersed in it. A moving fluid, however, also exerts tangential shear forces on the surface because of the no-slip condition caused by viscous effects. Both of these forces, in general, have components in the direction of flow, and thus the drag force is due to the combined effects of pressure and wall shear forces in the flow direction. The components of the pressure and wall shear forces in the *normal* direction to flow tend to move the body in that direction, and their sum is called **lift**.

In general, both the skin friction (wall shear) and pressure contribute to the drag and the lift. In the special case of a thin, flat plate aligned parallel to the flow direction, the drag force depends on the wall shear only and is independent of pressure. When the flat plate is placed normal to the flow direction, however, the drag force depends on the pressure only and is independent of the wall shear since the shear stress in this case acts in the direction normal to flow (Fig. 7–3). For slender bodies such as wings, the shear force acts nearly parallel to the flow direction. The drag force for such slender bodies is mostly due to shear forces (the skin friction).

The drag force F_D depends on the density ρ of the fluid, the upstream velocity V, and the size, shape, and orientation of the body, among other things. The drag characteristics of a body are represented by the dimensionless **drag coefficient** C_D defined as

Drag coefficient:
$$C_D = \frac{F_D}{\frac{1}{2}\rho V^2 A} \qquad \textbf{(7–1)}$$

where A is the *frontal area* (the area projected on a plane normal to the direction of flow) for blunt bodies—bodies that tend to block the flow. The frontal area of a cylinder of diameter D and length L, for example, is $A = LD$. For parallel flow over flat plates or thin airfoils, A is the surface area. The drag coefficient is primarily a function of the shape of the body, but it may also depend on the Reynolds number and the surface roughness.

The drag force is the net force exerted by a fluid on a body in the direction of flow due to the combined effects of wall shear and pressure forces. The part of drag that is due directly to wall shear stress τ_w is called the **skin friction drag** (or just *friction drag*) since it is caused by frictional effects, and the part that is due directly to pressure P is called the **pressure drag** (also called the *form drag* because of its strong dependence on the form or shape of the body). When the friction and pressure drag coefficients are available, the total drag coefficient is determined by simply adding them,

$$C_D = C_{D,\text{ friction}} + C_{D,\text{ pressure}} \qquad \textbf{(7–2)}$$

The *friction drag* is the component of the wall shear force in the direction of flow, and thus it depends on the orientation of the body as well as the magnitude of the wall shear stress τ_w. The friction drag is *zero* for a surface normal to flow and *maximum* for a surface parallel to flow since the friction drag in this case equals the total shear force on the surface. Therefore, for parallel flow over a flat plate, the drag coefficient is equal to the *friction drag coefficient,* or simply the *friction coefficient* (Fig. 7–4). That is,

Flat plate:
$$C_D = C_{D,\text{ friction}} = C_f \qquad \textbf{(7–3)}$$

Once the average friction coefficient C_f is available, the drag (or friction) force over the surface can be determined from Eq. 7–1. In this case, A is the surface area of the plate exposed to fluid flow. When both sides of a thin plate are subjected to flow, A becomes the total area of the top and bottom surfaces. Note that the friction coefficient, in general, varies with location along the surface.

Friction drag is a strong function of viscosity, and an "idealized" fluid with zero viscosity would produce zero friction drag since the wall shear stress would be zero. The pressure drag would also be zero in this case during steady flow regardless of the shape of the body since there are no pressure losses. For flow in the horizontal direction, for example, the pressure along a horizontal

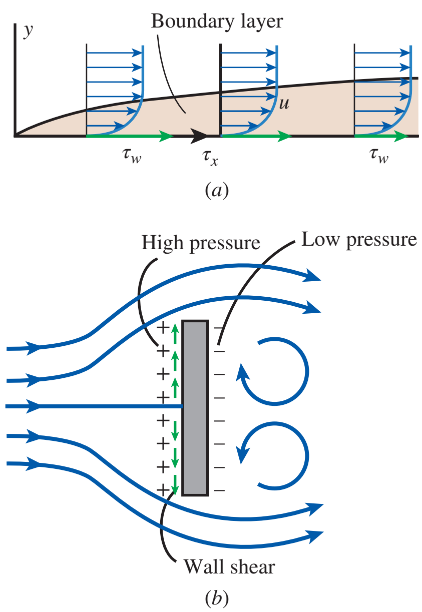

FIGURE 7–3
(*a*) Drag force acting on a flat plate parallel to the flow depends on wall shear only. (*b*) Drag force acting on a flat plate normal to the flow depends on the pressure only and is independent of the wall shear, which acts normal to the free-stream flow.

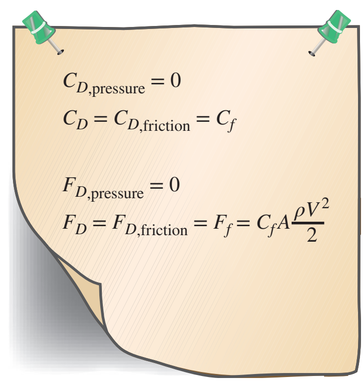

FIGURE 7–4
For parallel flow over a flat plate, the pressure drag is zero, and thus the drag coefficient is equal to the friction coefficient, and the drag force is equal to the friction force.

line is constant (just like stationary fluids) since the upstream velocity is constant, and thus there is no net pressure force acting on the body in the horizontal direction. Therefore, the total drag is zero for the case of ideal inviscid fluid flow.

At low Reynolds numbers, most drag is due to friction drag. This is especially the case for highly streamlined bodies such as airfoils. The friction drag is also proportional to the surface area. Therefore, bodies with a larger surface area experience a larger friction drag. Large commercial airplanes, for example, reduce their total surface area and thus drag by retracting their wing extensions when they reach the cruising altitudes to save fuel. The friction drag coefficient is independent of *surface roughness* in laminar flow, but it is a strong function of surface roughness in turbulent flow due to surface roughness elements protruding further into the boundary layer.

The pressure drag is proportional to the frontal area and to the *difference* between the pressures acting on the front and back of the immersed body. Therefore, the pressure drag is usually dominant for blunt bodies, negligible for streamlined bodies such as airfoils, and zero for thin, flat plates parallel to the flow.

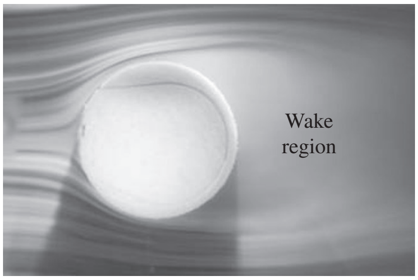

FIGURE 7–5
Separation during flow over a tennis ball and the wake region.
Courtesy NASA Ames Research Center and Cislunar Aerospace, Inc.

When a fluid separates from a body, it forms a separated region between the body and the fluid stream. This low-pressure region behind the body where recirculating and backflows occur is called the **separated region**. The larger the separated region, the larger the pressure drag. The effects of flow separation are felt far downstream in the form of reduced velocity (relative to the upstream velocity). The region of flow trailing the body where the effects of the body on velocity are felt is called the **wake** (Fig. 7–5). The separated region comes to an end when the two flow streams reattach. Therefore, the separated region is an enclosed volume, whereas the wake keeps growing behind the body until the fluid in the wake region regains its velocity and the velocity profile becomes nearly flat again. Viscous and rotational effects are the most significant in the boundary layer, the separated region, and the wake.

Heat Transfer

The phenomena that affect drag force also affect heat transfer, and this effect appears in the Nusselt number. By nondimensionalizing the boundary layer equations, it was shown in Chap. 6 that the local and average Nusselt numbers have the functional form

$$\mathrm{Nu}_x = f_1(x^*, \mathrm{Re}_x, \mathrm{Pr}) \quad \text{and} \quad \mathrm{Nu} = f_2(\mathrm{Re}_L, \mathrm{Pr}) \qquad \textbf{(7–4}a, b\textbf{)}$$

The experimental data for heat transfer are often represented conveniently with reasonable accuracy by a simple power-law relation of the form

$$\mathrm{Nu} = C\,\mathrm{Re}_L^m\,\mathrm{Pr}^n \qquad \textbf{(7–5)}$$

where m and n are constant exponents, and the value of the constant C depends on geometry and flow.

The fluid temperature in the thermal boundary layer varies from T_s at the surface to about T_∞ at the outer edge of the boundary. The fluid properties also vary with temperature, and thus with position across the boundary layer. In order to account for the variation of the properties with temperature, the fluid properties are usually evaluated at the so-called **film temperature**, defined as

$$T_f = \frac{T_s + T_\infty}{2} \quad \textbf{(7–6)}$$

which is the *arithmetic average* of the surface and the free-stream temperatures. The fluid properties are then assumed to remain constant at those values during the entire flow. An alternative way of accounting for the variation of properties with temperature is to evaluate all properties at the free-stream temperature and to multiply the Nusselt number relation in Eq. 7–5 by $(\mathrm{Pr}_\infty/\mathrm{Pr}_s)^r$ or $(\mu_\infty/\mu_s)^r$ where r is an experimentally determined constant.

The local drag and convection coefficients vary along the surface as a result of the changes in the velocity boundary layers in the flow direction. We are usually interested in the drag force and the heat transfer rate for the *entire* surface, which can be determined using the *average* friction and convection coefficient. Therefore, we present correlations for both local (identified with the subscript x) and average friction and convection coefficients. When relations for local friction and convection coefficients are available, the *average* friction and convection coefficients for the entire surface can be determined by integration from

$$C_f = \frac{1}{L}\int_0^L C_{f,x}\,dx \quad \textbf{(7–7)}$$

and

$$h = \frac{1}{L}\int_0^L h_x\,dx \quad \textbf{(7–8)}$$

When the average drag and convection coefficients are available, the drag force can be determined from Eq. 7–1, and the rate of heat transfer to or from an isothermal surface can be determined from

$$\dot{Q} = hA_s(T_s - T_\infty) \quad \textbf{(7–9)}$$

where A_s is the surface area.

7–2 ▪ PARALLEL FLOW OVER FLAT PLATES

Consider the parallel flow of a fluid over a flat plate of length L in the flow direction, as shown in Fig. 7–6. The x-coordinate is measured along the plate surface from the leading edge in the direction of the flow. The fluid approaches the plate in the x-direction with a uniform velocity V and temperature T_∞. The flow in the velocity boundary layers starts out as laminar, but if the plate is sufficiently long, the flow becomes turbulent at a distance x_{cr} from the leading edge where the Reynolds number reaches its critical value for transition.

The transition from laminar to turbulent flow depends on the *surface geometry, surface roughness, upstream velocity, surface temperature,* and the *type of fluid,* among other things, and it is best characterized by the Reynolds number.

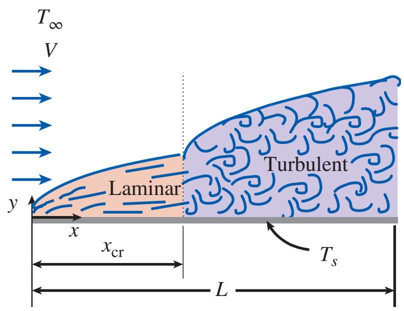

FIGURE 7–6
Laminar and turbulent regions of the boundary layer during flow over a flat plate.

The Reynolds number at a distance x from the leading edge of a flat plate is expressed as

$$\mathrm{Re}_x = \frac{\rho V x}{\mu} = \frac{Vx}{\nu} \tag{7–10}$$

Note that the value of the Reynolds number varies for a flat plate along the flow, reaching $\mathrm{Re}_L = VL/\nu$ at the end of the plate.

For flow over a flat plate, the transition from laminar to turbulent begins at about $\mathrm{Re} \cong 1 \times 10^5$, but it does not become fully turbulent before the Reynolds number reaches much higher values, typically around 3×10^6. In engineering analysis, a generally accepted value for the critical Reynolds number is

$$\mathrm{Re}_{\mathrm{cr}} = \frac{\rho V x_{\mathrm{cr}}}{\mu} = 5 \times 10^5 \tag{7–11}$$

The actual value of the engineering critical Reynolds number for a flat plate may vary somewhat from 10^5 to 3×10^6, depending on the surface roughness, the turbulence level, and the variation of pressure along the surface.

Friction Coefficient

Based on analysis, the boundary layer thickness and the local friction coefficient at location x for laminar flow over a flat plate were determined in Chap. 6 to be

$$\textit{Laminar:} \quad \delta = \frac{4.91x}{\mathrm{Re}_x^{1/2}} \quad \text{and} \quad C_{f,x} = \frac{0.664}{\mathrm{Re}_x^{1/2}}, \quad \mathrm{Re}_x < 5 \times 10^5 \tag{7–12a, b}$$

From experiments, the corresponding relations for turbulent flow are

$$\textit{Turbulent:} \quad \delta = \frac{0.38x}{\mathrm{Re}_x^{1/5}} \quad \text{and} \quad C_{f,x} = \frac{0.059}{\mathrm{Re}_x^{1/5}}, \quad 5 \times 10^5 \le \mathrm{Re}_x \le 10^7 \tag{7–13a, b}$$

where x is the distance from the leading edge of the plate and $\mathrm{Re}_x = Vx/\nu$ is the Reynolds number at location x. Note that $C_{f,x}$ is proportional to $\mathrm{Re}_x^{-1/2}$ and thus to $x^{-1/2}$ for laminar flow. Therefore, $C_{f,x}$ is supposedly *infinite* at the leading edge ($x = 0$) and decreases by a factor of $x^{-1/2}$ in the flow direction. The local friction coefficients are higher in turbulent flow than they are in laminar flow because of the intense mixing that occurs in the turbulent boundary layer. Note that $C_{f,x}$ reaches its highest values when the flow becomes fully turbulent, and it then decreases by a factor of $x^{-1/5}$ in the flow direction.

The *average* friction coefficient over the entire plate is determined by substituting the preceding relations into Eq. 7–7 and performing the integrations (Fig. 7–7). We get

$$\textit{Laminar:} \quad C_f = \frac{1.33}{\mathrm{Re}_L^{1/2}} \quad \mathrm{Re}_L < 5 \times 10^5 \tag{7–14}$$

$$\textit{Turbulent:} \quad C_f = \frac{0.074}{\mathrm{Re}_L^{1/5}} \quad 5 \times 10^5 \le \mathrm{Re}_L \le 10^7 \tag{7–15}$$

The first relation gives the average friction coefficient for the entire plate when the flow is *laminar* over the *entire* plate. Note that the average friction coefficient over the entire plate in case of laminar flow is twice the value of the local friction coefficient at the end of the plate, $C_f = 2C_{f,x=L}$. The second relation gives the average friction coefficient for the entire plate only when

$$\begin{aligned}
C_f &= \frac{1}{L}\int_0^L C_{f,x}\,dx \\
&= \frac{1}{L}\int_0^L \frac{0.664}{\mathrm{Re}_x^{1/2}}\,dx \\
&= \frac{0.664}{L}\int_0^L \left(\frac{Vx}{\nu}\right)^{-1/2} dx \\
&= \frac{0.664}{L}\left(\frac{V}{\nu}\right)^{-1/2} \left.\frac{x^{1/2}}{\frac{1}{2}}\right|_0^L \\
&= \frac{2 \times 0.664}{L}\left(\frac{V}{\nu L}\right)^{-1/2} \\
&= \frac{1.33}{\mathrm{Re}_L^{1/2}}
\end{aligned}$$

FIGURE 7–7
The average friction coefficient over a surface is determined by integrating the local friction coefficient over the entire surface. The values shown here are for a laminar flat plate boundary layer.

the flow is *turbulent* over the *entire* plate, or when the laminar flow region of the plate is too small relative to the turbulent flow region (that is, $x_{cr} \ll L$).

In some cases, a flat plate is sufficiently long for the flow to become turbulent, but not long enough to disregard the laminar flow region. In such cases, the *average* friction coefficient over the entire plate is determined by performing the integration in Eq. 7–7 over two parts: use Eq. 7–12*b* for the laminar region $0 \le x \le x_{cr}$ and Eq. 7–13*b* for the turbulent region $x_{cr} < x \le L$ as

$$C_f = \frac{1}{L}\left(\int_0^{x_{cr}} C_{f,\,x\,\text{laminar}}\,dx + \int_{x_{cr}}^{L} C_{f,\,x,\,\text{turbulent}}\,dx\right) \qquad \textbf{(7–16)}$$

Note that we included the transition region with the turbulent region. Again taking the critical Reynolds number to be $\text{Re}_{cr} = 5 \times 10^5$ and performing the integrations of Eq. 7–16 after substituting the indicated expressions, the *average* friction coefficient over the entire plate is determined to be

$$C_f = \frac{0.074}{\text{Re}_L^{1/5}} - \frac{1742}{\text{Re}_L} \qquad 5 \times 10^5 \le \text{Re}_L \le 10^7 \qquad \textbf{(7–17)}$$

For a completely turbulent boundary layer ($\text{Re}_{cr} = 0$) or a very short x_{cr} ($L \gg x_{cr}$ or $\text{Re}_L \gg \text{Re}_{cr}$), Eq. 7–17 simplifies to the equation for turbulent flow, Eq. 7–15. The constants in Eq. 7–17 will be different for different critical Reynolds numbers. Also, the surfaces are assumed to be *smooth* and the free stream to be *turbulence free*. For laminar flow, the friction coefficient depends on only the Reynolds number, and the surface roughness has no effect. For turbulent flow, however, surface roughness causes the friction coefficient to increase severalfold, to the point that in fully turbulent regime the friction coefficient is a function of surface roughness alone and is independent of the Reynolds number (Fig. 7–8). This is also the case in pipe flow.

A curve fit of experimental data for the average friction coefficient in this regime is given by Schlichting (1979) as

$$\textit{Rough surface, turbulent:} \quad C_f = \left(1.89 - 1.62\log\frac{\varepsilon}{L}\right)^{-2.5} \qquad \textbf{(7–18)}$$

where ε is the surface roughness and L is the length of the plate in the flow direction. In the absence of a better relation, the relation above can be used for turbulent flow on rough surfaces for $\text{Re} > 10^6$, especially when $\varepsilon/L > 10^{-4}$.

Relative roughness, ε/L	Friction coefficient C_f
0.0*	0.0029
1×10^{-5}	0.0032
1×10^{-4}	0.0049
1×10^{-3}	0.0084

*Smooth surface for $\text{Re} = 10^7$. Others calculated from Eq. 7–18 for fully rough flow.

FIGURE 7–8
For turbulent flow, surface roughness may cause the friction coefficient to increase severalfold.

Heat Transfer Coefficient

The local Nusselt number at a location x for laminar flow over a flat plate for the case of constant wall temperature was determined in Chap. 6 by solving the differential energy equation to be

$$\textit{Laminar:} \quad \text{Nu}_x = \frac{h_x x}{k} = 0.332\,\text{Re}_x^{0.5}\,\text{Pr}^{1/3} \qquad \text{Pr} > 0.6, \quad \text{Re}_x < 5 \times 10^5 \qquad \textbf{(7–19)}$$

From experiments, the corresponding relation for turbulent flow is

$$\textit{Turbulent:} \quad \text{Nu}_x = \frac{h_x x}{k} = 0.0296\,\text{Re}_x^{0.8}\,\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5 \times 10^5 \le \text{Re}_x \le 10^7 \end{array} \qquad \textbf{(7–20)}$$

Note that h_x is proportional to $\text{Re}_x^{0.5}$ and thus to $x^{-0.5}$ for laminar flow. Therefore, h_x is *infinite* at the leading edge ($x = 0$) and decreases by a factor of $x^{-0.5}$ in the

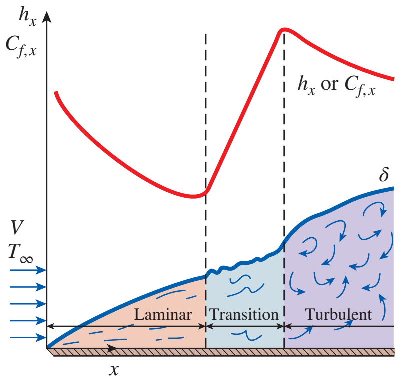

FIGURE 7–9
The variation of the local friction and heat transfer coefficients for flow over a flat plate.

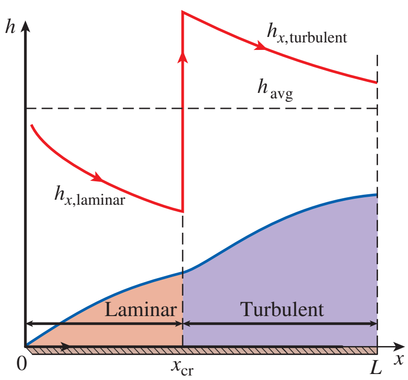

FIGURE 7–10
Graphical representation of the average heat transfer coefficient for a flat plate with combined laminar and turbulent flow.

flow direction. The variation of the boundary layer thickness δ and the friction and heat transfer coefficients along an isothermal flat plate are shown in Fig. 7–9. The local friction and heat transfer coefficients are higher in turbulent flow than they are in laminar flow. Also, h_x reaches its highest values when the flow becomes fully turbulent, and then decreases by a factor of $x^{-0.2}$ in the flow direction, as shown in the figure.

The *average* Nusselt number over the entire plate is determined by substituting the relations above into Eq. 7–8 and performing the integrations. We get

$$\textit{Laminar:}\quad \text{Nu} = \frac{hL}{k} = 0.664\ \text{Re}_L^{0.5}\ \text{Pr}^{1/3} \qquad \text{Re}_L < 5\times10^5,\ \text{Pr} > 0.6 \tag{7–21}$$

$$\textit{Turbulent:}\quad \text{Nu} = \frac{hL}{k} = 0.037\ \text{Re}_L^{0.8}\ \text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5\times10^5 \le \text{Re}_L \le 10^7 \end{array} \tag{7–22}$$

The first relation gives the average heat transfer coefficient for the entire plate when the flow is *laminar* over the *entire* plate. Note that the average Nusselt number over the entire plate in case of laminar flow is twice the value of the local Nusselt number at the end of the plate, $\text{Nu} = 2\text{Nu}_{x=L}$ or $h = 2h_{x=L}$. This is only true for laminar flow and does not carry over to the turbulent flow. The second relation gives the average heat transfer coefficient for the entire plate only when the flow is *turbulent* over the *entire* plate, or when the laminar flow region of the plate is too small relative to the turbulent flow region.

In some cases, a flat plate is sufficiently long for the flow to become turbulent, but not long enough to disregard the laminar flow region. In such cases, the *average* heat transfer coefficient over the entire plate is determined by performing the integration in Eq. 7–8 over two parts: use Eq. 7–19 for the laminar region $0 \le x \le x_{\text{cr}}$ and Eq. 7–20 for the turbulent region $x_{\text{cr}} \le x \le L$ as

$$h = \frac{1}{L}\left(\int_0^{x_{\text{cr}}} h_{x,\ \text{laminar}}\, dx + \int_{x_{\text{cr}}}^{L} h_{x,\ \text{turbulent}}\, dx\right) \tag{7–23}$$

Again taking the critical Reynolds number to be $\text{Re}_{\text{cr}} = 5\times10^5$ and performing the integrations in Eq. 7–23 after substituting the indicated expressions, the *average* Nusselt number over the *entire* plate is determined to be (Fig. 7–10)

$$\text{Nu} = \frac{hL}{k} = (0.037\ \text{Re}_L^{0.8} - 871)\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5\times10^5 \le \text{Re}_L \le 10^7 \end{array} \tag{7–24}$$

For a completely turbulent boundary layer ($\text{Re}_{\text{cr}} = 0$) or a very short x_{cr}($L \gg x_{\text{cr}}$ or $\text{Re}_L \gg \text{Re}_{\text{cr}}$), Eq. 7–24 simplifies to the equation for turbulent flow, Eq. 7–22. The constants in Eq. 7–24 will be different for different critical Reynolds numbers.

Liquid metals such as mercury have high thermal conductivities, and they are commonly used in applications that require high heat transfer rates. Liquid metals have very small Prandtl numbers, and Eq. 7–19 does not apply. However, for this case the thermal boundary layer develops much faster than the velocity boundary layer. Then we can assume the velocity in the thermal boundary layer to be constant at the free stream value and solve the energy equation. It gives

$$\text{Nu}_x = 0.565(\text{Re}_x\,\text{Pr})^{1/2} = 0.565\ \text{Pe}_x^{1/2} \qquad \text{Pr} \le 0.05,\ \text{Pe}_x \ge 100 \tag{7–25}$$

where $\text{Pe}_x = \text{Re}_x\text{Pr}$ is the dimensionless **Peclet number** (Fig. 7–11).

It is desirable to have a single correlation that applies to *all fluids*, including liquid metals. By curve-fitting existing data, Churchill and Ozoe (1973) proposed the following relation for laminar flow over an isothermal flat plate which is applicable for *all Prandtl numbers* and is claimed to be accurate to ±1 percent,

$$\mathrm{Nu}_x = \frac{h_x x}{k} = \frac{0.3387\,\mathrm{Pr}^{1/3}\,\mathrm{Re}_x^{1/2}}{[1 + (0.0468/\mathrm{Pr})^{2/3}]^{1/4}} \qquad \mathrm{Re}_x\,\mathrm{Pr} \geq 100 \tag{7–26}$$

with $\mathrm{Nu} = 2\mathrm{Nu}_{x=L}$. For the *uniform heat flux* case, 0.3387 is changed to 0.4637 and 0.0468 is changed to 0.02052. Properties are still evaluated at the film temperature.

These relations have been obtained for the case of *isothermal* surfaces, but they could also be used approximately for the case of nonisothermal surfaces by assuming the surface temperature to be constant at some average value. Also, the surfaces are assumed to be *smooth,* and the free stream to be *turbulent free*. The effect of variable properties can be accounted for by evaluating all properties at the film temperature.

FIGURE 7–11
Jean Claude Eugene Peclet (1793–1857), a French physicist, was born in Besancon, France. He was one of the first scholars of the Ecole Normale at Paris. His publications were famous for their clarity of style, sharp-minded views, and well-performed experiments. The dimensionless **Peclet number** is named after him.
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Flat Plate with Unheated Starting Length

So far we have limited our consideration to situations for which the entire plate is heated from the leading edge. But many practical applications involve surfaces with an unheated starting section of length ξ, shown in Fig. 7–12, and thus there is no heat transfer for $0 < x < \xi$. In such cases, the velocity boundary layer starts to develop at the leading edge ($x = 0$), but the thermal boundary layer starts to develop where heating starts ($x = \xi$).

Consider a flat plate whose heated section is maintained at a constant temperature ($T = T_s$ constant for $x > \xi$). Using integral solution methods, the local Nusselt numbers for both laminar and turbulent flows are determined to be

Laminar:
$$\mathrm{Nu}_x = \frac{\mathrm{Nu}_{x\,(\text{for }\xi=0)}}{[1 - (\xi/x)^{3/4}]^{1/3}} = \frac{0.332\,\mathrm{Re}_x^{0.5}\,\mathrm{Pr}^{1/3}}{[1 - (\xi/x)^{3/4}]^{1/3}} \tag{7–27}$$

Turbulent:
$$\mathrm{Nu}_x = \frac{\mathrm{Nu}_{x\,(\text{for }\xi=0)}}{[1 - (\xi/x)^{9/10}]^{1/9}} = \frac{0.0296\,\mathrm{Re}_x^{0.8}\,\mathrm{Pr}^{1/3}}{[1 - (\xi/x)^{9/10}]^{1/9}} \tag{7–28}$$

for $x > \xi$. Note that for $\xi = 0$, these Nu_x relations reduce to $\mathrm{Nu}_{x(\text{for }\xi=0)}$, which is the Nusselt number relation for a flat plate without an unheated starting length. Therefore, the terms in brackets in the denominator serve as correction factors for plates with unheated starting lengths.

The determination of the average Nusselt number for the heated section of a plate requires the integration of the local Nusselt number relations above, which cannot be done analytically. Therefore, integrations must be done numerically. The results of numerical integrations have been correlated for the average Nusselt number (Ameel, 1997) as

Laminar:
$$\mathrm{Nu} = \mathrm{Nu}_{(\text{for }\xi=0)}\frac{L}{L-\xi}[1 - (\xi/L)^{3/4}]^{2/3} \tag{7–29}$$

Turbulent:
$$\mathrm{Nu} = \mathrm{Nu}_{(\text{for }\xi=0)}\frac{L}{L-\xi}[1 - (\xi/L)^{9/10}]^{8/9} \tag{7–30}$$

The quantity $\mathrm{Nu}_{(\text{for }\xi=0)}$ is the average Nusselt number for a plate of length L when heating starts at the leading edge of the plate. For laminar flow,

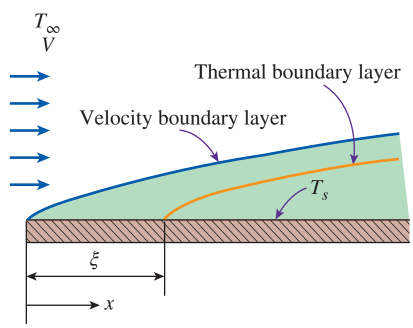

FIGURE 7–12
Flow over a flat plate with an unheated starting length.

it can be obtained from Eq. 7–21; for turbulent flow, it is obtained from Eq. 7–22 (assuming turbulent flow over the entire surface). Note that Nu is equal to hL/k, where h is averaged over the heated portion of the plate only, which has the length $(L - \xi)$. To determine the total rate of heat transfer from the plate, the averaged value of h over the heated portion of the plate must be multiplied by the area of the heated section.

Uniform Heat Flux

When a flat plate is subjected to *uniform heat flux* instead of uniform temperature, the local Nusselt number is given by

Laminar: $$\text{Nu}_x = 0.453\,\text{Re}_x^{0.5}\,\text{Pr}^{1/3} \qquad \text{Pr} > 0.6, \qquad \text{Re}_x < 5 \times 10^5 \tag{7–31}$$

Turbulent: $$\text{Nu}_x = 0.0308\,\text{Re}_x^{0.8}\,\text{Pr}^{1/3} \qquad 0.6 \le \text{Pr} \le 60, \qquad 5 \times 10^5 \le \text{Re}_x \le 10^7 \tag{7–32}$$

These relations give values that are 36 percent higher for laminar flow and 4 percent higher for turbulent flow relative to the isothermal plate case. When the plate involves an unheated starting length, the relations developed for the uniform surface temperature case can still be used provided that Eqs. 7–31 and 7–32 are used for $\text{Nu}_{x(\text{for }\xi=0)}$ in Eqs. 7–27 and 7–28, respectively. To obtain the *average* Nusselt number for a flat plate subjected to *uniform heat flux*, it is sufficiently accurate to use the average Nusselt number equations developed for the case of isothermal surfaces for laminar flow (Eq. 7–21) and turbulent flow (Eq. 7–22).

When heat flux $\dot{q}_s$ is prescribed, the rate of heat transfer to or from the plate and the surface temperature at a distance x are determined from

$$\dot{Q} = \dot{q}_s A_s \tag{7–33}$$

and

$$\dot{q}_s = h_x[T_s(x) - T_\infty] \rightarrow T_s(x) = T_\infty + \frac{\dot{q}_s}{h_x} \tag{7–34}$$

where A_s is the heat transfer surface area.

General Solutions for Simultaneously Moving Plates and Fluids

The results presented so far in this chapter primarily correspond to the case of fluid passing over a stationary plate. However, in some circumstances, both the plate and the fluid are in motion. For example, in the thermal processing of sheetlike material such as paper, linoleum, polymeric sheets, roofing shingles, insulating materials, and fine-fiber matts, the sheet moves parallel to its own plane. The moving sheet may induce motion in the neighboring fluid or, alternatively, the fluid may have an independent forced convection motion that is parallel to that of the sheet. The generalized solutions for this scenario are available by correlations developed in Sparrow and Abraham (2005).

EXAMPLE 7–1 Flow of Hot Oil Over a Flat Plate

Engine oil at 60°C flows over the upper surface of a 5-m-long flat plate whose temperature is 20°C with a velocity of 2 m/s (Fig. 7–13). Determine the total drag force and the rate of heat transfer per unit width of the entire plate.

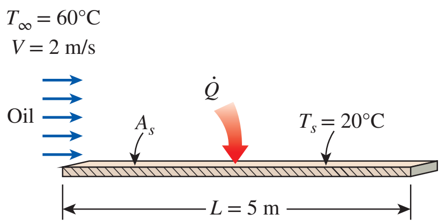

FIGURE 7–13
Schematic for Example 7–1.

SOLUTION Engine oil flows over a flat plate. The total drag force and the rate of heat transfer per unit width of the plate are to be determined.

Assumptions **1** The flow is steady and incompressible. **2** The critical Reynolds number is $Re_{cr} = 5 \times 10^5$.

Properties The properties of engine oil at the film temperature of $T_f = (T_s + T_\infty)/2 = (20 + 60)/2 = 40°C$ are (Table A–13)

$$\rho = 876 \text{ kg/m}^3 \qquad \text{Pr} = 2962$$
$$k = 0.1444 \text{ W/m·K} \qquad \nu = 2.485 \times 10^{-4} \text{ m}^2\text{/s}$$

Analysis Noting that $L = 5$ m, the Reynolds number at the end of the plate is

$$\text{Re}_L = \frac{VL}{\nu} = \frac{(2 \text{ m/s})(5 \text{ m})}{2.485 \times 10^{-4} \text{ m}^2\text{/s}} = 4.024 \times 10^4$$

which is less than the critical Reynolds number. Thus we have *laminar flow* over the entire plate, and the average friction coefficient is

$$C_f = 1.33 \text{ Re}_L^{-0.5} = 1.33 \times (4.024 \times 10^4)^{-0.5} = 0.00663$$

Noting that the pressure drag is zero and thus $C_D = C_f$ for parallel flow over a flat plate, the drag force acting on the plate per unit width becomes

$$F_D = C_f A \frac{\rho V^2}{2} = 0.00663(5 \times 1 \text{ m}^2)\frac{(876 \text{ kg/m}^3)(2 \text{ m/s})^2}{2}\left(\frac{1 \text{ N}}{1 \text{ kg·m/s}^2}\right) = \mathbf{58.1 \text{ N}}$$

The total drag force acting on the entire plate can be determined by multiplying the value obtained above by the width of the plate.

This force per unit width corresponds to the weight of a mass of about 6 kg. Therefore, a person who applies an equal and opposite force to the plate to keep it from moving will feel like he or she is using as much force as is necessary to hold a 6-kg mass from dropping.

Similarly, the Nusselt number is determined using the laminar flow relations for a flat plate,

$$\text{Nu} = \frac{hL}{k} = 0.664 \text{ Re}_L^{0.5} \text{ Pr}^{1/3} = 0.664 \times (4.024 \times 10^4)^{0.5} \times 2962^{1/3} = 1913$$

Then,

$$h = \frac{k}{L}\text{Nu} = \frac{0.144 \text{ W/m·K}}{5 \text{ m}}(1913) = 55.25 \text{ W/m}^2\text{·K}$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (55.25 \text{ W/m}^2\text{·K})(5 \times 1 \text{ m}^2)(60 - 20)°\text{C} = \mathbf{11{,}050 \text{ W}}$$

Discussion Note that heat transfer is always from the higher-temperature medium to the lower-temperature one. In this case, it is from the oil to the plate. The heat transfer rate is per meter width of the plate. The heat transfer for the entire plate can be obtained by multiplying the value obtained by the actual width of the plate.

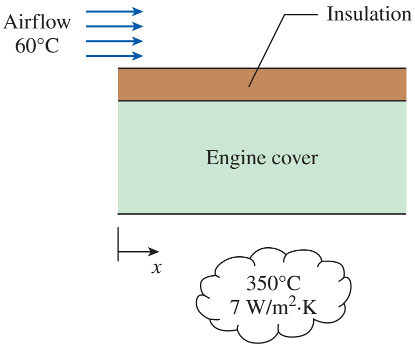

FIGURE 7–14
Schematic for Example 7–2.

PtD **EXAMPLE 7–2** **Prevention of Fire Hazard in the Event of Oil Leakage**

Heat dissipated from an engine in operation can cause hot spots on its surface. If the outer surface of an engine is situated in a place where oil leakage is possible, then when leaked oil comes in contact with hot spots above the oil's autoignition temperature, it can ignite spontaneously. Consider an engine cover that is made of a stainless steel plate with a thickness of 1 cm and a thermal conductivity of 14 W/m·K. The stainless steel plate is covered with a 5-mm-thick insulation ($k = 0.5$ W/m·K). The inner surface of the engine cover is exposed to hot air at 350°C with a convection heat transfer coefficient of 7 W/m^2·K (Fig. 7–14). The 2-m-long engine outer surface is cooled by air blowing in parallel over it at 7 m/s in an environment where the ambient air is at 60°C. To prevent fire hazard in the event of oil leak on the engine cover, the engine cover surface should be kept below 180°C. It has been determined that the 5-mm-thick insulation layer is not sufficient to keep the engine cover surface below 180°C. To solve this problem, one of the plant supervisors suggested adjusting the blower capacity to provide an increase in the cooling air velocity by 10 percent. Determine if this is a viable method for keeping the engine cover surface below 180°C. Evaluate the air properties at 120°C.

SOLUTION In this example, the concepts of prevention through design (PtD) are applied in conjunction with external forced convection and thermal resistance analysis. An engine cover with a layer of insulation is subjected to convection heat transfer on the inner and outer surfaces. To prevent a fire hazard by keeping the engine outer surface temperature below 180°C, it is suggested to increase the cooling air velocity by 10 percent. The effectiveness of this method is to be evaluated.

Assumptions **1** The thermal properties of the plate and insulation are constant. **2** One-dimensional heat conduction through the plate. **3** Uniform plate surface temperature. **4** Thermal contact resistance at interface is negligible. **5** Radiation effects are negligible. **6** Local atmospheric pressure is 1 atm. **7** The critical Reynolds number is $\text{Re}_{cr} = 5 \times 10^5$.

Properties The thermal conductivities of the stainless steel and the insulation are given to be $k_{ss} = 14$ W/m·K and $k_{ins} = 0.5$ W/m·K, respectively. The properties of air are evaluated at $T_f = 120$°C: $k = 0.03235$ W/m·K, $\nu = 2.522 \times 10^{-5}$ m^2/s, and Pr = 0.7073 (from Table A–15).

Analysis When the cooling air velocity is increased by 10 percent (7.7 m/s), the Reynolds number for the 2-m-long plate is

$$\text{Re}_L = \frac{VL}{\nu} = \frac{(7.7 \text{ m/s})(2 \text{ m})}{2.522 \times 10^{-5} \text{ m}^2/\text{s}} = 610{,}626 > 5 \times 10^5$$

With the Reynolds number between $5 \times 10^5 < \text{Re}_L < 10^7$, the proper equation is the combined laminar and turbulent relation for the Nusselt number:

$$\begin{aligned} \text{Nu} &= \frac{hL}{k} = (0.037\,\text{Re}_L^{0.8} - 871)\text{Pr}^{1/3} \\ &= [0.037(610{,}626)^{0.8} - 871](0.7073)^{1/3} \\ &= 625.77 \end{aligned}$$

The convection heat transfer coefficient on the engine outer surface is

$$h = \text{Nu}\frac{k}{L} = 625.77\left(\frac{0.03235 \text{ W/m·K}}{2 \text{ m}}\right) = 10.122 \text{ W/m}^2\text{·K}$$

From Chap. 3, the thermal resistances of different layers are

$$R_{\text{conv},\,i} = \frac{1}{h_i A} \text{(inside surface convection resistance)},$$

$$R_{\text{ss}} = \frac{L_{\text{ss}}}{k_{\text{ss}} A} \text{(stainless steel layer resistance)},$$

$$R_{\text{ins}} = \frac{L_{\text{ins}}}{k_{\text{ins}} A} \text{(insulation layer resistance)},$$

$$R_{\text{conv},\,o} = \frac{1}{h_o A} \text{(outside surface convection resistance)}$$

Then,

$$A R_{\text{total}} = A(R_{\text{conv},\,i} + R_{\text{ss}} + R_{\text{ins}} + R_{\text{conv},\,o}) = \frac{1}{h_i} + \frac{L_{\text{ss}}}{k_{\text{ss}}} + \frac{L_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_o}$$

$$= \frac{1}{7\ \text{W/m}^2\cdot\text{K}} + \frac{0.01\ \text{m}}{14\ \text{W/m}\cdot\text{K}} + \frac{0.005\ \text{m}}{0.5\ \text{W/m}\cdot\text{K}} + \frac{1}{10.122\ \text{W/m}^2\cdot\text{K}}$$

$$= 0.25237\ \text{m}^2\cdot\text{K/W}$$

and

$$AR_{\text{conv},\,o} = \frac{1}{h_o} = \frac{1}{10.122\ \text{W/m}^2\cdot\text{K}} = 0.09879\ \text{m}^2\cdot\text{K/W}$$

The heat flux through the layers is

$$\dot{q} = \frac{\dot{Q}}{A} = \frac{T_{\infty,\,i} - T_{\infty,\,o}}{AR_{\text{total}}} = \frac{T_{s,\,o} - T_{\infty,\,o}}{AR_{\text{conv},\,o}} \rightarrow T_{s,\,o} = \frac{R_{\text{conv},\,o}}{R_{\text{total}}}(T_{\infty,\,i} - T_{\infty,\,o}) + T_{\infty,\,o}$$

The outer surface temperature is

$$T_{s,\,o} = \frac{0.09879\ \text{m}^2\cdot\text{K/W}}{0.25237\ \text{m}^2\cdot\text{K/W}}(350 - 60)^\circ\text{C} + 60^\circ\text{C} = \mathbf{173.5^\circ C}$$

Yes, the suggested method is a viable method.

Discussion Increasing the cooling air velocity from 7 m/s to 7.7 m/s (by only 10 percent), kept the engine outer surface temperature below the fire hazard limit of 180°C. Another method that could be explored would be to increase the insulation thickness.

EXAMPLE 7–3 **Cooling of Plastic Sheets by Forced Air**

The forming section of a plastics plant puts out a continuous sheet of plastic that is 4 ft wide and 0.04 in thick at a velocity of 30 ft/min. The temperature of the plastic sheet is 200°F when it is exposed to the surrounding air, and a 2-ft-long section of the plastic sheet is subjected to airflow at 80°F at a velocity of 10 ft/s on both sides along its surfaces normal to the direction of motion of the sheet, as shown in Fig. 7–15. Determine (*a*) the rate of heat transfer from the plastic sheet to air by forced convection and radiation and (*b*) the temperature of the plastic sheet at the end of the cooling section. Take the density, specific heat, and emissivity of the plastic sheet to be $\rho = 75\ \text{lbm/ft}^3$, $c_p = 0.4\ \text{Btu/lbm}\cdot{}^\circ\text{F}$, and $\varepsilon = 0.9$.

SOLUTION Plastic sheets are cooled as they leave the forming section of a plastics plant. The rate of heat loss from the plastic sheet by convection and radiation and the exit temperature of the plastic sheet are to be determined.

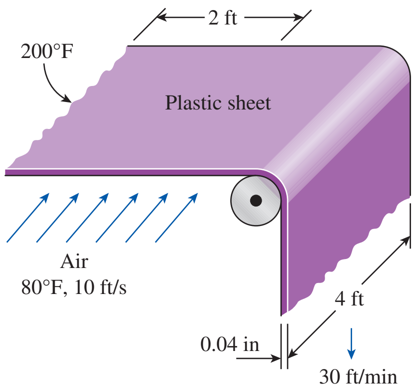

FIGURE 7–15
Schematic for Example 7–3.

Assumptions **1** Steady operating conditions exist. **2** The critical Reynolds number is $\text{Re}_{cr} = 5 \times 10^5$. **3** Air is an ideal gas. **4** The local atmospheric pressure is 1 atm. **5** The surrounding surfaces are at the temperature of the room air.

Properties The properties of the plastic sheet are given in the problem statement. The properties of air at the film temperature of $T_f = (T_s + T_\infty)/2 = (200 + 80)/2 = 140°\text{F}$ and 1 atm pressure are (Table A–15E)

$$k = 0.01623 \text{ Btu/h·ft·°F} \qquad \text{Pr} = 0.7202$$
$$\nu = 0.204 \times 10^{-3} \text{ ft}^2\text{/s}$$

Analysis (*a*) We expect the temperature of the plastic sheet to drop somewhat as it flows through the 2-ft-long cooling section, but at this point we do not know the magnitude of that drop. Therefore, we assume the plastic sheet to be isothermal at 200°F to get started. We will repeat the calculations if necessary to account for the temperature drop of the plastic sheet.

Noting that $L = 4$ ft, the Reynolds number at the end of the airflow across the plastic sheet is

$$\text{Re}_L = \frac{VL}{\nu} = \frac{(10 \text{ ft/s})(4 \text{ ft})}{0.204 \times 10^{-3} \text{ ft}^2\text{/s}} = 1.961 \times 10^5$$

which is less than the critical Reynolds number. Thus, we have *laminar flow* over the entire sheet, and the Nusselt number is determined from the laminar flow relations for a flat plate to be

$$\text{Nu} = \frac{hL}{k} = 0.664 \text{ Re}_L^{0.5} \text{ Pr}^{1/3} = 0.664 \times (1.961 \times 10^5)^{0.5} \times (0.7202)^{1/3} = 263.6$$

Then,

$$h = \frac{k}{L}\text{Nu} = \frac{0.01623 \text{ Btu/h·ft·°F}}{4 \text{ ft}}(263.6) = 1.07 \text{ Btu/h·ft}^2\text{·°F}$$
$$A_s = (2 \text{ ft})(4 \text{ ft})(2 \text{ sides}) = 16 \text{ ft}^2$$

and

$$\begin{aligned}
\dot{Q}_{\text{conv}} &= hA_s(T_s - T_\infty) \\
&= (1.07 \text{ Btu/h·ft}^2\text{·°F})(16 \text{ ft}^2)(200 - 80)\text{°F} \\
&= 2054 \text{ Btu/h} \\
\dot{Q}_{\text{rad}} &= \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4) \\
&= (0.9)(0.1714 \times 10^{-8} \text{ Btu/h·ft}^2\text{·R}^4)(16 \text{ ft}^2)[(660 \text{ R})^4 - (540 \text{ R})^4] \\
&= 2585 \text{ Btu/h}
\end{aligned}$$

Therefore, the rate of cooling of the plastic sheet by combined convection and radiation is

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = 2054 + 2585 = \mathbf{4639 \text{ Btu/h}}$$

(*b*) To find the temperature of the plastic sheet at the end of the cooling section, we need to know the mass of the plastic rolling out per unit time (or the mass flow rate), which is determined from

$$\dot{m} = \rho A_c V_{\text{plastic}} = (75 \text{ lbm/ft}^3)\left(\frac{4 \times 0.04}{12}\text{ft}^2\right)\left(\frac{30}{60}\text{ft/s}\right) = 0.5 \text{ lbm/s}$$

Then, an energy balance on the cooled section of the plastic sheet yields

$$\dot{Q} = \dot{m}c_p(T_2 - T_1) \rightarrow T_2 = T_1 + \frac{\dot{Q}}{\dot{m}c_p}$$

Noting that $\dot{Q}$ is a negative quantity (heat loss) for the plastic sheet and substituting, the temperature of the plastic sheet as it leaves the cooling section is determined to be

$$T_2 = 200°\text{F} + \frac{-4639 \text{ Btu/h}}{(0.5 \text{ lbm/s})(0.4 \text{ Btu/lbm·°F})}\left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = \mathbf{193.6°F}$$

Discussion The average temperature of the plastic sheet drops by about 6.4°F as it passes through the cooling section. The calculations now can be repeated by taking the average temperature of the plastic sheet to be 196.8°F instead of 200°F for better accuracy, but the change in the results will be insignificant because of the small change in temperature.

EXAMPLE 7–4 Heat Transfer from a Flat Plate with Unheated Starting Length

An airstream at 1 atm flows, with a velocity of 4 m/s, in parallel over a 2-m-long flat plate where there is an unheated starting length of 0.5 m. The airstream has a temperature of 20°C, and the heated section of the flat plate is maintained at a constant temperature of 80°C. Determine (*a*) the average Nusselt number and convection heat transfer coefficient for the heated section and (b) the rate of heat transfer per unit width for the heated section.

SOLUTION An airstream flows in parallel over a flat plate where there is an unheated starting length. The average Nusselt number, the average convection heat transfer coefficient, and the rate of heat transfer per unit width for the heated section are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Surface temperature is uniform throughout the heated section. **3** Thermal properties are constant. **4** The critical Reynolds number is $\text{Re}_{cr} = 5 \times 10^5$.

Properties The properties of air at the film temperature of $T_f = (T_s + T_\infty)/2 = (80 + 20)/2 = 50°\text{C}$ and 1 atm pressure are (Table A–15E)

$$k = 0.02735 \text{ W/m·K} \qquad \text{Pr} = 0.7228 \qquad \nu = 1.798 \times 10^{-5} \text{ m}^2\text{/s}$$

Analysis (*a*) The Reynolds number at $L = 2$ m is

$$\text{Re}_L = \frac{VL}{\nu} = \frac{(4 \text{ m/s})(2 \text{ m})}{1.798 \times 10^{-5} \text{ m}^2\text{/s}} = 4.449 \times 10^5$$

Since $\text{Re}_L < 5 \times 10^5$, the flow over the entire plate is laminar. Using the proper relation for Nusselt number, the average Nusselt number over the heated section can be determined:

$$\text{Nu} = \text{Nu}_{(\text{for } \xi=0)}\frac{L}{L-\xi}[1 - (\xi/L)^{3/4}]^{2/3}$$

where

$$\text{Nu}_{(\text{for } \xi=0)} = 0.664 \text{ Re}_L^{0.5} \text{ Pr}^{1/3} = 0.664(4.449 \times 10^5)^{0.5}(0.7228)^{1/3} = 397.5$$

then

$$\text{Nu} = (397.5)\frac{2 \text{ m}}{2 \text{ m} - 0.5 \text{ m}}[1 - (0.5 \text{ m/2 m})^{3/4}]^{2/3} = \mathbf{396.2}$$

The average convection heat transfer coefficient over the heated section is

$$h = \text{Nu}\frac{k}{L} = 396.2\left(\frac{0.02735 \text{ W/m·K}}{2 \text{ m}}\right) = \mathbf{5.42 \text{ W/m}^2\text{·K}}$$

(*b*) The rate of heat transfer per unit width (w) for the heated section is

$$\dot{Q} = hA(T_s - T_\infty) = h(L - \xi)w(T_s - T_\infty)$$
$$\dot{Q}/w = (5.42 \text{ W/m}^2\cdot\text{K})(2 \text{ m} - 0.5 \text{ m})(80 - 20)\text{K} = \mathbf{487.8 \text{ W/m}}$$

Discussion The heat transfer in this case is from the surface to the airstream (from the higher temperature to the lower temperature). Once the actual width of the plate is known, the heat transfer for the entire plate can be determined. For the plate with unheated starting length, the thermal boundary layer does not begin to grow until the heated section, while the velocity boundary layer begins at the leading edge.

7–3 ▪ FLOW ACROSS CYLINDERS AND SPHERES

Flow across cylinders and spheres is frequently encountered in practice. For example, the tubes in a shell-and-tube heat exchanger involve both *internal flow* through the tubes and *external flow* over the tubes, and both flows must be considered in the analysis of the heat exchanger. Also, many sports such as soccer, tennis, and golf involve flow over spherical balls.

The characteristic length for a circular cylinder or sphere is taken to be the *external diameter D*. Thus, the Reynolds number is defined as $\text{Re} = \text{Re}_D = VD/\nu$ where V is the uniform velocity of the fluid as it approaches the cylinder or sphere. The critical Reynolds number for flow across a circular cylinder or sphere is about $\text{Re}_{cr} \cong 2 \times 10^5$. That is, the boundary layer remains laminar for about $\text{Re} \lesssim 2 \times 10^5$ and becomes turbulent for $\text{Re} \gtrsim 2 \times 10^5$.

Crossflow over a cylinder exhibits complex flow patterns, as shown in Fig. 7–16. The fluid approaching the cylinder branches out and encircles the cylinder, forming a boundary layer that wraps around the cylinder. The fluid particles on the midplane strike the cylinder at the stagnation point, bringing the fluid to a complete stop and thus raising the pressure at that point. The pressure decreases in the flow direction while the fluid velocity increases.

At very low upstream velocities ($\text{Re} \lesssim 1$), the fluid completely wraps around the cylinder, and the two arms of the fluid meet on the rear side of the cylinder in an orderly manner. Thus, the fluid follows the curvature of the cylinder. At higher velocities, the fluid still hugs the cylinder on the frontal side, but it is too fast to remain attached to the surface as it approaches the top (or bottom) of the cylinder. As a result, the boundary layer detaches from the surface, forming a separation region behind the cylinder. Flow in the wake region is characterized by periodic vortex formation and pressures much lower than the stagnation point pressure.

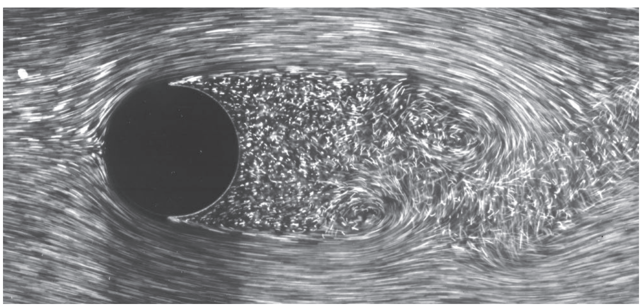

FIGURE 7–16
Laminar boundary layer separation with a turbulent wake; flow over a circular cylinder at Re = 2000.
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The nature of the flow across a cylinder or sphere strongly affects the total drag coefficient C_D. Both the *friction drag* and the *pressure drag* can be significant. The high pressure in the vicinity of the stagnation point and the low pressure on the opposite side in the wake produce a net force on the body in the direction of flow. The drag force is primarily due to friction drag at low Reynolds numbers (Re < 10) and to pressure drag at high Reynolds numbers (Re > 5000). Both effects are significant at intermediate Reynolds numbers.

From dimensional analysis it can be shown that the average drag coefficients C_D for a smooth single circular cylinder and a sphere is a function of Reynolds number, $C_D = f(\text{Re}_D)$, as shown in Fig. 7–17. The curves exhibit different behaviors in different ranges of Reynolds numbers:

- For $\text{Re} \lesssim 1$, we have creeping flow, and the drag coefficient decreases with increasing Reynolds number. For a sphere, it is $C_D = 24/\text{Re}$. There is no flow separation in this regime.
- At about Re = 10, separation starts occurring on the rear of the body with vortex shedding starting at about $\text{Re} \cong 90$. The region of separation increases with increasing Reynolds number up to about $\text{Re} = 10^3$. At this point, the drag is mostly (about 95 percent) due to pressure drag. The drag coefficient continues to decrease with increasing Reynolds number in this range of $10 < \text{Re} < 10^3$. (A decrease in the drag coefficient does not necessarily indicate a decrease in drag. The drag force is proportional to the square of the velocity, and the increase in velocity at higher Reynolds numbers usually more than offsets the decrease in the drag coefficient.)
- In the moderate range of $10^3 < \text{Re} < 10^5$, the drag coefficient remains relatively constant. This behavior is characteristic of blunt bodies. The flow in the boundary layer is laminar in this range, but the flow in the separated region past the cylinder or sphere is highly turbulent, with a wide, turbulent wake.
- There is a sudden drop in the drag coefficient somewhere in the range of $10^5 < \text{Re} < 10^6$ (usually, at about 2×10^5). This large reduction in C_D is due to the flow in the boundary layer becoming *turbulent,* which moves the separation point further toward the rear of the body, reducing the size

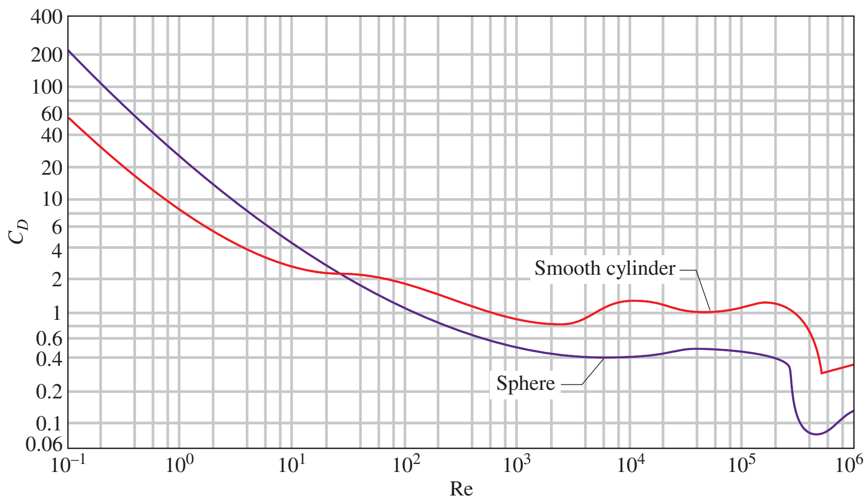
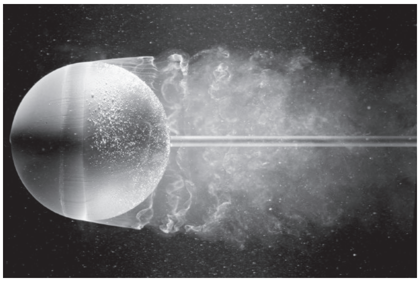
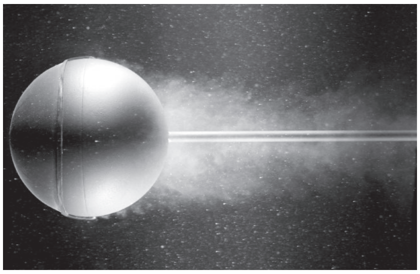


FIGURE 7–17
Average drag coefficient for crossflow over a smooth circular cylinder and a smooth sphere.
Source: H. Schlichting, Boundary Layer Theory 7e. *Copyright ©1979 The McGraw-Hill Companies, Inc.*



(*a*)



(*b*)

FIGURE 7–18
Flow visualization of flow over (*a*) a smooth sphere at Re = 15,000 and (*b*) a sphere at Re = 30,000 with a trip wire. The delay of boundary layer separation is clearly seen by comparing the two photographs.
©Henri Werlé—ONERA, the French Aerospace Lab

of the wake and thus the magnitude of the pressure drag. This is in contrast to streamlined bodies, which experience an increase in the drag coefficient (mostly due to friction drag) when the boundary layer becomes turbulent.

Flow separation occurs at about $\theta \cong 80°$ (measured from the front stagnation point of a cylinder) when the boundary layer is *laminar* and at about $\theta \cong 140°$ when it is *turbulent* (Fig. 7–18). The delay of separation in turbulent flow is caused by the rapid fluctuations of the fluid in the transverse direction, which enable the turbulent boundary layer to travel farther along the surface before separation occurs, resulting in a narrower wake and a smaller pressure drag. Keep in mind that turbulent flow has a fuller velocity profile than in the laminar case, and thus it requires a stronger adverse pressure gradient to overcome the additional momentum close to the wall. In the range of Reynolds numbers where the flow changes from laminar to turbulent, even the drag force F_D decreases as the velocity (and thus the Reynolds number) increases. This results in a sudden decrease in drag of a flying body (sometimes called the *drag crisis*) and instabilities in flight.

Effect of Surface Roughness

We mentioned earlier that *surface roughness,* in general, increases the drag coefficient in turbulent flow. This is especially the case for streamlined bodies. For blunt bodies such as a circular cylinder or sphere, however, an increase in the surface roughness may actually *decrease* the drag coefficient, as shown in Fig. 7–19 for a sphere. This is done by tripping the boundary layer into turbulence at a lower Reynolds number and thus causing the fluid to close in behind the body, narrowing the wake and reducing pressure drag considerably. This results in a much smaller drag coefficient and thus drag force for a rough-surfaced cylinder or sphere in a certain range of Reynolds number compared to a smooth one of identical size at the same velocity. At Re = 2×10^5, for example, $C_D \cong 0.1$ for a rough sphere with $\varepsilon/D = 0.0015$, whereas $C_D \cong 0.5$ for a smooth one. Therefore, the drag coefficient in this case is reduced by a factor of 5 by simply roughening the surface. Note, however, that

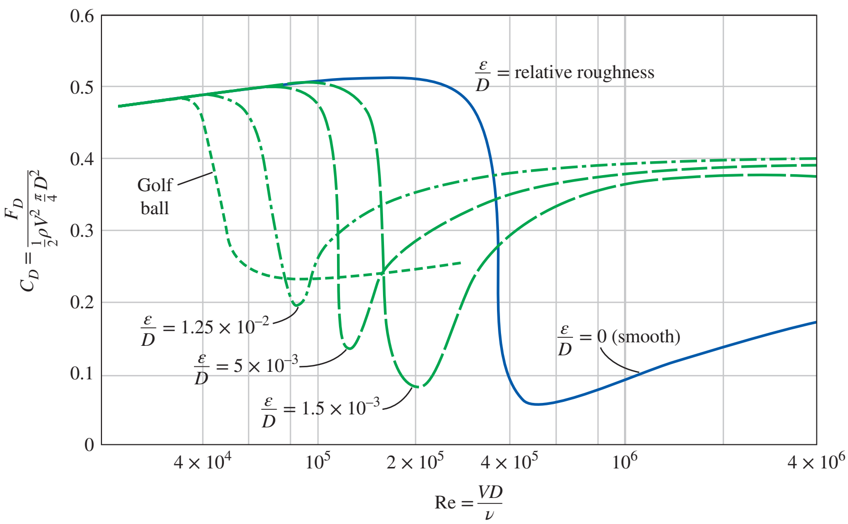


FIGURE 7–19
The effect of surface roughness on the drag coefficient of a sphere.
Source: Blevins, 1984.

at Re $= 10^6$, $C_D \cong 0.4$ for a very rough sphere while $C_D \cong 0.1$ for the smooth one. Obviously, roughening the sphere in this case will increase the drag by a factor of 4 (Fig. 7–20).

The preceding discussion shows that roughening the surface can be used to great advantage in reducing drag, but it can also backfire on us if we are not careful—specifically, if we do not operate in the right range of the Reynolds number. With this consideration, golf balls are intentionally roughened to induce *turbulence* at a lower Reynolds number to take advantage of the sharp *drop* in the drag coefficient at the onset of turbulence in the boundary layer (the typical velocity range of golf balls is 15 to 150 m/s, and the Reynolds number is less than 4×10^5). The critical Reynolds number of dimpled golf balls is about 4×10^4. The occurrence of turbulent flow at this Reynolds number reduces the drag coefficient of a golf ball by about half, as shown in Fig. 7–19. For a given hit, this means a longer distance for the ball. Experienced golfers also give the ball a spin during the hit, which helps the rough ball develop a lift and thus travel higher and farther. A similar argument can be given for a tennis ball. For a table tennis ball, however, the distances are very short, and the balls never reach the speeds in the turbulent range. Therefore, the surfaces of table tennis balls are made smooth.

Once the drag coefficient is available, the drag force acting on a body in crossflow can be determined from Eq. 7–1 where A is the *frontal area* ($A = LD$ for a cylinder of length L and $A = \pi D^2/4$ for a sphere). It should be kept in mind that free-stream turbulence and disturbances by other bodies in the flow (such as flow over tube bundles) may affect the drag coefficients significantly.

	C_D	
Re	Smooth Surface	Rough Surface, $\varepsilon/D = 0.0015$
2×10^5	0.5	0.1
10^6	0.1	0.4

FIGURE 7–20
Surface roughness may increase or decrease the drag coefficient of a spherical object, depending on the value of the Reynolds number.

EXAMPLE 7–5 Drag Force Acting on a Pipe in a River

A 2.2-cm-outer-diameter pipe is to span a river at a 30-m-wide section while being completely immersed in water (Fig. 7–21). The average flow velocity of water is 4 m/s, and the water temperature is 15°C. Determine the drag force exerted on the pipe by the river.

SOLUTION A pipe is submerged in a river. The drag force that acts on the pipe is to be determined.

Assumptions **1** The outer surface of the pipe is smooth so that Fig. 7–17 can be used to determine the drag coefficient. **2** Water flow in the river is steady. **3** The direction of water flow is normal to the pipe. **4** Turbulence in river flow is not considered.

Properties The density and dynamic viscosity of water at 15°C are $\rho = 999.1$ kg/m^3 and $\mu = 1.138 \times 10^{-3}$ kg/m·s (Table A–9).

Analysis Noting that $D = 0.022$ m, the Reynolds number is

$$\text{Re} = \frac{VD}{\nu} = \frac{\rho VD}{\mu} = \frac{(999.1 \text{ kg/m}^3)(4 \text{ m/s})(0.022 \text{ m})}{1.138 \times 10^{-3} \text{ kg/m·s}} = 7.73 \times 10^4$$

The drag coefficient corresponding to this value is, from Fig. 7–17, $C_D = 1.0$. Also, the frontal area for flow past a cylinder is $A = LD$. Then the drag force acting on the pipe becomes

$$F_D = C_D A\frac{\rho V^2}{2} = 1.0(30 \times 0.022 \text{ m}^2)\frac{(999.1 \text{ kg/m}^3)(4 \text{ m/s})^2}{2}\left(\frac{1 \text{ N}}{1 \text{ kg·m/s}^2}\right)$$
$$= 5275 \text{ N} \cong \mathbf{5.30 \text{ kN}}$$

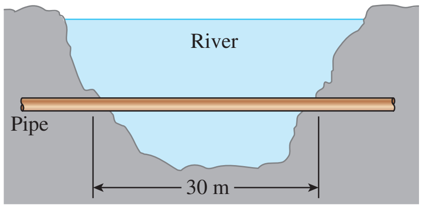


FIGURE 7–21
Schematic for Example 7–5.

Discussion Note that this force is equivalent to the weight of a mass over 500 kg. Therefore, the drag force the river exerts on the pipe is equivalent to hanging a total of over 500 kg in mass on the pipe supported at its ends 30 m apart. The necessary precautions should be taken if the pipe cannot support this force. If the river were to flow at a faster speed or if turbulent fluctuations in the river were more significant, the drag force would be even larger. *Unsteady* forces on the pipe might then be significant.

Reading C_D values from Fig. 7–17 for a smooth cylinder in crossflow could at times be difficult. Hooman and Dukhan (2013) proposed a simple correlation, $C_D = 1 + (6/\text{Re})$ for the Reynolds number range of $10^2 < \text{Re} < 10^5$.

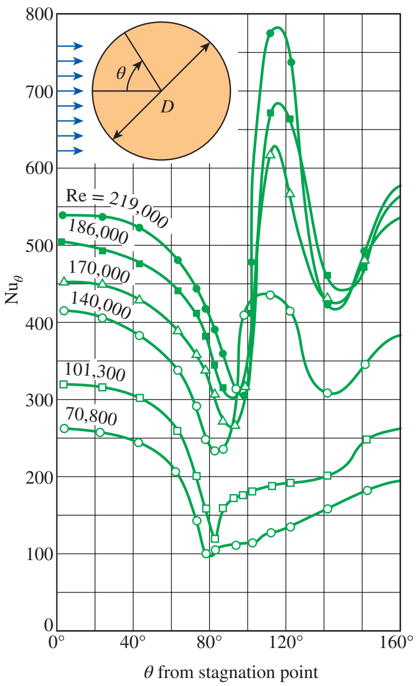


FIGURE 7–22
Variation of the local heat transfer coefficient along the circumference of a circular cylinder in crossflow of air.
Source: Giedt, 1949.

Heat Transfer Coefficient

Flows across cylinders and spheres, in general, involve *flow separation,* which is difficult to handle analytically. Therefore, such flows must be studied experimentally or numerically. Indeed, flow across cylinders and spheres has been studied experimentally by many investigators, and several empirical correlations have been developed for the heat transfer coefficient.

The complicated flow pattern across a cylinder greatly influences heat transfer. The variation of the local Nusselt number Nu_θ around the periphery of a cylinder subjected to crossflow of air is given in Fig. 7–22. Note that, for all cases, the value of Nu_θ starts out relatively high at the stagnation point ($\theta = 0°$) but decreases with increasing θ as a result of the thickening of the laminar boundary layer. On the two curves at the bottom corresponding to Re = 70,800 and 101,300, Nu_θ reaches a minimum at $\theta \approx 80°$, which is the separation point in laminar flow. Then Nu_θ increases with increasing θ as a result of the intense mixing in the separated flow region (the wake). The curves at the top corresponding to Re = 140,000 to 219,000 differ from the first two curves in that they have *two* minima for Nu_θ. The sharp increase in Nu_θ at about $\theta \approx 90°$ is due to the transition from laminar to turbulent flow. The later decrease in Nu_θ is again due to the thickening of the boundary layer. Nu_θ reaches its second minimum at about $\theta \approx 140°$, which is the flow separation point in turbulent flow, and increases with θ as a result of the intense mixing in the turbulent wake region.

The preceding discussions on the local heat transfer coefficients are insightful; however, they are of limited value in heat transfer calculations since the calculation of heat transfer requires the *average* heat transfer coefficient over the entire surface. Of the several such relations available in the literature for the average Nusselt number for crossflow over a cylinder, we present the one proposed by Churchill and Bernstein (1977):

$$\text{Nu}_{\text{cyl}} = \frac{hD}{k} = 0.3 + \frac{0.62\,\text{Re}^{1/2}\,\text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}}\left[1 + \left(\frac{\text{Re}}{282{,}000}\right)^{5/8}\right]^{4/5} \tag{7–35}$$

This relation is quite comprehensive in that it correlates available data well for RePr > 0.2. The fluid properties are evaluated at the *film temperature* $T_f = \frac{1}{2}(T_\infty + T_s)$, which is the average of the free-stream and surface temperatures.

For flow over a *sphere,* Whitaker (1972) recommends the following comprehensive correlation:

$$\text{Nu}_{\text{sph}} = \frac{hD}{k} = 2 + [0.4\,\text{Re}^{1/2} + 0.06\,\text{Re}^{2/3}]\text{Pr}^{0.4}\left(\frac{\mu_\infty}{\mu_s}\right)^{1/4} \tag{7–36}$$

which is valid for $3.5 \le \text{Re} \le 8 \times 10^4$, $0.7 \le \text{Pr} \le 380$ and $1.0 \le (\mu_\infty/\mu_s) \le 3.2$. The fluid properties in this case are evaluated at the free-stream temperature

T_∞, except for μ_s, which is evaluated at the surface temperature T_s. Although the two relations above are considered to be quite accurate, the results obtained from them can be off by as much as 30 percent.

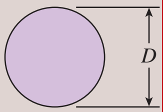
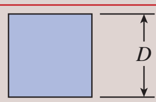
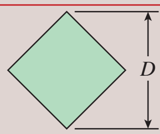
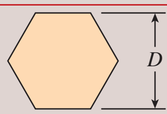
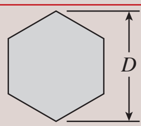
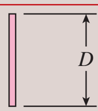
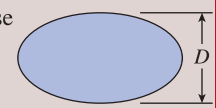
The average Nusselt number for flow across cylinders can be expressed compactly as

$$\text{Nu}_{\text{cyl}} = \frac{hD}{k} = C\,\text{Re}^m\,\text{Pr}^n \tag{7–37}$$

where $n = \frac{1}{3}$ and the experimentally determined constants C and m are given in Table 7–1 for circular as well as various noncircular cylinders. The characteristic length D for use in the calculation of the Reynolds and the Nusselt numbers for different geometries is as indicated on the figure. All fluid properties are evaluated at the film temperature. Note that the values presented in Table 7–1 for noncircular geometries have been updated based on the recommendations of Sparrow et al. (2004). Correlations for square and chamfered cylinders for all possible directions of incoming fluid are available in Tong et al. (2016).

TABLE 7–1

Empirical correlations for the average Nusselt number for forced convection over circular and noncircular cylinders in crossflow

Cross-Section of the Cylinder	Fluid	Range of Re	Nusselt Number
Circle 	Gas or liquid	0.4–4 4–40 40–4000 4000–40,000 40,000–400,000	$\text{Nu} = 0.989\,\text{Re}^{0.330}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.911\,\text{Re}^{0.385}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.683\,\text{Re}^{0.466}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.193\,\text{Re}^{0.618}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.027\,\text{Re}^{0.805}\,\text{Pr}^{1/3}$
Square 	Gas	3900–79,000 32,000–1,600,000	$\text{Nu} = 0.094\,\text{Re}^{0.675}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.0249\,\text{Re}^{0.811}\,\text{Pr}^{1/3}$
Square (tilted 45°) 	Gas	5600–111,000 46,000–2,300,000	$\text{Nu} = 0.258\,\text{Re}^{0.588}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.0260\,\text{Re}^{0.839}\,\text{Pr}^{1/3}$
Hexagon 	Gas	4500–90,700	$\text{Nu} = 0.148\,\text{Re}^{0.638}\,\text{Pr}^{1/3}$
Hexagon (tilted 45°) 	Gas	5200–20,400 20,400–105,000	$\text{Nu} = 0.162\,\text{Re}^{0.638}\,\text{Pr}^{1/3}$ $\text{Nu} = 0.039\,\text{Re}^{0.782}\,\text{Pr}^{1/3}$
Vertical plate 	Gas	6300–23,600	$\text{Nu} = 0.257\,\text{Re}^{0.731}\,\text{Pr}^{1/3}$
Ellipse 	Gas	1400–8200	$\text{Nu} = 0.197\,\text{Re}^{0.612}\,\text{Pr}^{1/3}$

Source: Zukauskas, 1972, Jakob 1949, Sparrow et al., 2004, and Tong et al., 2016.

The relations for cylinders presented here are for *single* cylinders or cylinders oriented such that the flow over them is not affected by the presence of others. Also, they are applicable to *smooth* surfaces. *Surface roughness* and the *free-stream turbulence* may affect the drag and heat transfer coefficients significantly. Eq. 7–37 provides a simpler alternative to Eq. 7–35 for flow over cylinders. However, Eq. 7–35 is more accurate, and thus it should be preferred in calculations whenever possible.

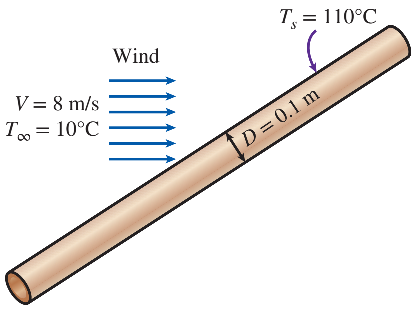


FIGURE 7–23
Schematic for Example 7–6.

EXAMPLE 7–6 Heat Loss from a Steam Pipe in Windy Air

A long 10-cm-diameter steam pipe whose external surface temperature is 110°C passes through some open area that is not protected against the winds (Fig. 7–23). Determine the rate of heat loss from the pipe per unit of its length when the air is at 1 atm pressure and 10°C and the wind is blowing across the pipe at a velocity of 8 m/s.

SOLUTION A steam pipe is exposed to windy air. The rate of heat loss from the steam is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Radiation effects are negligible. **3** Air is an ideal gas.

Properties The properties of air at the average film temperature of $T_f = (T_s + T_\infty)/2 = (110 + 10)/2 = 60°C$ and 1 atm pressure are (Table A–15)

$$k = 0.02808 \text{ W/m·K} \qquad \Pr = 0.7202 \qquad \nu = 1.896 \times 10^{-5} \text{ m}^2\text{/s}$$

Analysis The Reynolds number is

$$\text{Re} = \frac{VD}{\nu} = \frac{(8 \text{ m/s})(0.1 \text{ m})}{1.896 \times 10^{-5} \text{ m}^2\text{/s}} = 4.219 \times 10^4$$

The Nusselt number can be determined from

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{ Re}^{1/2} \Pr^{1/3}}{[1 + (0.4/\Pr)^{2/3}]^{1/4}}\left[1 + \left(\frac{\text{Re}}{282{,}000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(4.219 \times 10^4)^{1/2}(0.7202)^{1/3}}{[1 + (0.4/0.7202)^{2/3}]^{1/4}}\left[1 + \left(\frac{4.219 \times 10^4}{282{,}000}\right)^{5/8}\right]^{4/5} \\ &= 124 \end{aligned}$$

and

$$h = \frac{k}{D}\text{Nu} = \frac{0.02808 \text{ W/m·K}}{0.1 \text{ m}}(124) = 34.8 \text{ W/m}^2\text{·K}$$

Then, the rate of heat transfer from the pipe per unit of its length becomes

$$A_s = pL = \pi DL = \pi(0.1 \text{ m})(1 \text{ m}) = 0.314 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (34.8 \text{ W/m}^2\text{·K})(0.314 \text{ m}^2)(110 - 10)°\text{C} = \mathbf{1093 \text{ W}}$$

The rate of heat loss from the entire pipe can be obtained by multiplying the value above by the length of the pipe in m.

Discussion The simpler Nusselt number relation in Table 7–1 in this case would give Nu = 128, which is 3 percent higher than the value obtained above using Eq. 7–35.

EXAMPLE 7–7 Cooling of a Steel Ball by Forced Air

A 25-cm-diameter stainless steel ball ($\rho = 8055$ kg/m^3, $c_p = 480$ J/kg·K) is removed from the oven at a uniform temperature of 300°C (Fig. 7–24). The ball is then subjected to the flow of air at 1 atm pressure and 25°C with a velocity of 3 m/s. The surface temperature of the ball eventually drops to 200°C. Determine the average convection heat transfer coefficient during this cooling process, and estimate how long the process will take.

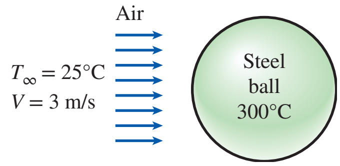


FIGURE 7–24
Schematic for Example 7–7.

SOLUTION A hot stainless steel ball is cooled by forced air. The average convection heat transfer coefficient and the cooling time are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Radiation effects are negligible. **3** Air is an ideal gas. **4** The outer surface temperature of the ball is uniform at all times. **5** The surface temperature of the ball during cooling is changing. Therefore, the convection heat transfer coefficient between the ball and the air will also change. To avoid this complexity, we take the surface temperature of the ball to be constant at the average temperature of (300 + 200)/2 = 250°C in the evaluation of the heat transfer coefficient and use the value obtained for the entire cooling process.

Properties The dynamic viscosity of air at the average surface temperature is $\mu_s = \mu_{@\,250°C} = 2.76 \times 10^{-5}$ kg/m·s. The properties of air at the free-stream temperature of 25°C and 1 atm are (Table A–15)

$$k = 0.02551 \text{ W/m·K} \qquad \nu = 1.562 \times 10^{-5} \text{ m}^2\text{/s}$$
$$\mu = 1.849 \times 10^{-5} \text{ kg/m·s} \qquad \text{Pr} = 0.7296$$

Analysis The Reynolds number is determined from

$$\text{Re} = \frac{VD}{\nu} = \frac{(3 \text{ m/s})(0.25 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2\text{/s}} = 4.802 \times 10^4$$

The Nusselt number is

$$\begin{aligned}\text{Nu} = \frac{hD}{k} &= 2 + [0.4\,\text{Re}^{1/2} + 0.06\,\text{Re}^{2/3}]\,\text{Pr}^{0.4}\left(\frac{\mu_\infty}{\mu_s}\right)^{1/4} \\ &= 2 + [0.4\,(4.802 \times 10^4)^{1/2} + 0.06\,(4.802 \times 10^4)^{2/3}](0.7296)^{0.4} \\ &\quad \times \left(\frac{1.849 \times 10^{-5}}{2.76 \times 10^{-5}}\right)^{1/4} \\ &= 135\end{aligned}$$

Then the average convection heat transfer coefficient becomes

$$h = \frac{k}{D}\text{Nu} = \frac{0.02551 \text{ W/m·K}}{0.25 \text{ m}}(135) = \mathbf{13.8\ W/m^2·K}$$

In order to estimate the time of cooling of the ball from 300°C to 200°C, we determine the *average* rate of heat transfer from Newton's law of cooling by using the *average* surface temperature. That is,

$$A_s = \pi D^2 = \pi(0.25 \text{ m})^2 = 0.1963 \text{ m}^2$$

$$\dot{Q}_{\text{avg}} = hA_s(T_{s,\,\text{avg}} - T_\infty) = (13.8 \text{ W/m}^2\text{·K})(0.1963 \text{ m}^2)(250 - 25)\text{°C} = 610 \text{ W}$$

Next we determine the *total* heat transferred from the ball, which is simply the change in the energy of the ball as it cools from 300°C to 200°C:

$$m = \rho V = \rho \tfrac{1}{6}\pi D^3 = (8055 \text{ kg/m}^3)\ \tfrac{1}{6}\pi(0.25 \text{ m})^3 = 65.9 \text{ kg}$$

$$Q_{\text{total}} = mc_p(T_2 - T_1) = (65.9 \text{ kg})(480 \text{ J/kg·K})(300 - 200)\text{°C} = 3{,}163{,}000 \text{ J}$$

In this calculation, we assumed that the entire ball is at 200°C, which is not necessarily true. The inner region of the ball will probably be at a higher temperature than its surface. With this assumption, the time of cooling is determined to be

$$\Delta t \approx \frac{Q}{\dot{Q}_{\text{avg}}} = \frac{3{,}163{,}000 \text{ J}}{610 \text{ J/s}} = 5185 \text{ s} = \mathbf{1\ h\ 26\ min}$$

Discussion The time of cooling could also be determined more accurately using the relations introduced in Chap. 4. But the simplifying assumptions we made above can be justified if all we need is a ballpark value. It will be naive to expect the time of cooling to be exactly 1 h 26 min, but, using our engineering judgment, it is realistic to expect the time of cooling to be somewhere between one and two hours.

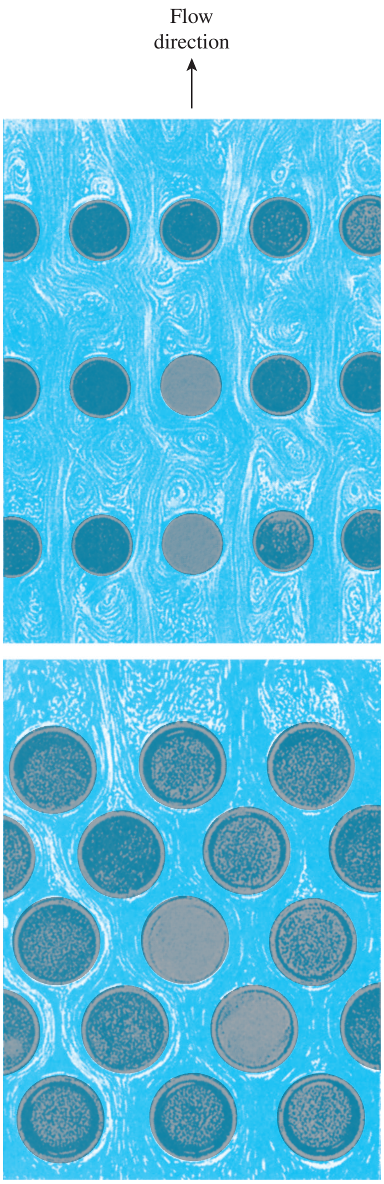


FIGURE 7–25
Flow patterns for in-line and staggered tube banks.

7–4 ▪ FLOW ACROSS TUBE BANKS

Crossflow over tube banks is commonly encountered in practice in heat transfer equipment such as the condensers and evaporators of power plants, refrigerators, and air conditioners. In such equipment, one fluid moves through the tubes while the other moves over the tubes in a perpendicular direction.

In a heat exchanger that involves a tube bank, the tubes are usually placed in a *shell* (and thus the name *shell-and-tube heat exchanger*), especially when the fluid is a liquid, and the fluid flows through the space between the tubes and the shell. There are many types of shell-and-tube heat exchangers, some of which are considered in Chap. 11. In this section we consider the general aspects of flow over a tube bank, and we try to develop a better and more intuitive understanding of the performance of heat exchangers that involve a tube bank.

Flow *through* the tubes can be analyzed by considering flow through a single tube and multiplying the results by the number of tubes. This is not the case for flow *over* the tubes, however, since the tubes affect the flow pattern and turbulence level downstream, and thus heat transfer to or from them, as shown in Fig. 7–25. Therefore, when analyzing heat transfer from a tube bank in crossflow, we must consider all the tubes in the bundle at once.

The tubes in a tube bank are usually arranged either *in-line* or *staggered* in the direction of flow, as shown in Fig. 7–26. The outer tube diameter D is taken as the characteristic length. The arrangement of the tubes in the tube bank is characterized by the *transverse pitch* S_T, the *longitudinal pitch* S_L, and the *diagonal pitch* S_D between tube centers. The diagonal pitch is determined from

$$S_D = \sqrt{S_L^2 + (S_T/2)^2} \qquad \textbf{(7–38)}$$

As the fluid enters the tube bank, the flow area decreases from $A_1 = S_T L$ to $A_T = (S_T - D)\,L$ between the tubes, and thus flow velocity increases. In staggered arrangement, the velocity may increase further in the diagonal region if the tube rows are very close to each other. In tube banks, the flow characteristics are dominated by the maximum velocity $V_{\max}$ that occurs within the tube bank rather than the approach velocity V. Therefore, the Reynolds number is defined on the basis of maximum velocity as

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{V_{\max} D}{\nu} \qquad \textbf{(7–39)}$$

The maximum velocity is determined from the conservation of mass requirement for steady incompressible flow. For *in-line* arrangement, the maximum velocity occurs at the minimum flow area between the tubes, and the conservation of mass can be expressed as (see Fig. 7–26*a*) $\rho V A_1 = \rho V_{\max} A_T$ or $V S_T = V_{\max}(S_T - D)$, Then the maximum velocity becomes

$$V_{\max} = \frac{S_T}{S_T - D} V \qquad \textbf{(7–40)}$$

In *staggered* arrangement, the fluid approaching through area A_1 in Fig. 7–26*b* passes through area A_T and then through area $2A_D$ as it wraps around the pipe in the next row. If $2A_D > A_T$, maximum velocity still occurs at A_T between the tubes, and thus the $V_{\max}$ relation Eq. 7–40 can also be used for staggered tube banks. But if $2A_D < A_T$ [or, if $2(S_D - D) < (S_T - D)$], maximum velocity occurs at the diagonal cross sections, and the maximum velocity in this case becomes

$$\textit{Staggered and } S_D < (S_T + D)/2: \qquad V_{\max} = \frac{S_T}{2(S_D - D)} V \qquad \textbf{(7–41)}$$

since $\rho V A_1 = \rho V_{\max}(2A_D)$ or $V S_T = 2V_{\max}(S_D - D)$.

The nature of flow around a tube in the first row resembles flow over a single tube discussed in Section 7–3, especially when the tubes are not too close to each other. Therefore, each tube in a tube bank that consists of a single transverse row can be treated as a single tube in crossflow. The nature of flow around a tube in the second and subsequent rows is very different, however, because of wakes formed and the turbulence caused by the tubes upstream. The level of turbulence, and thus the heat transfer coefficient, increases with row number because of the combined effects of upstream rows. But there is no significant change in turbulence level after the first few rows, and thus the heat transfer coefficient remains constant.

Flow through tube banks is studied experimentally since it is too complex to be treated analytically. We are primarily interested in the average heat transfer coefficient for the entire tube bank, which depends on the number of tube rows along the flow as well as the arrangement and the size of the tubes.

Several correlations, all based on experimental data, have been proposed for the average Nusselt number for crossflow over tube banks. More recently, Zukauskas (1987) has proposed correlations whose general form is

$$\text{Nu}_D = \frac{hD}{k} = C\,\text{Re}_D^m\,\text{Pr}^n\,(\text{Pr}/\text{Pr}_s)^{0.25} \qquad \textbf{(7–42)}$$

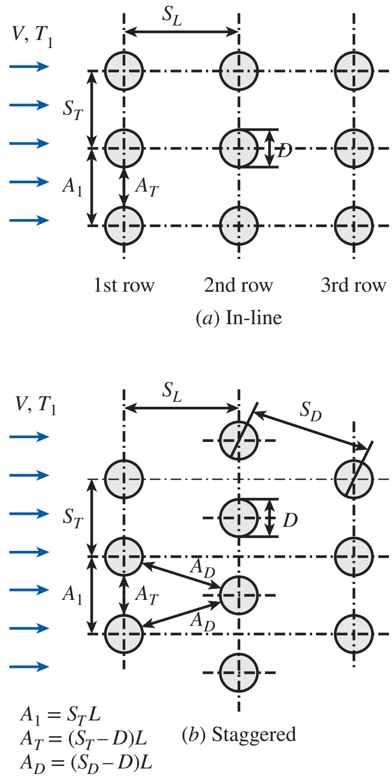


FIGURE 7–26
Arrangement of the tubes in in-line and staggered tube banks (A_1, A_T, and A_D are flow areas at indicated locations, and L is the length of the tubes).

where the values of the constants C, m, and n depend on Reynolds number. Such correlations are given in Table 7–2 for tube banks with more than 16 rows ($N_L > 16$), $0.7 < \text{Pr} < 500$ and $0 < \text{Re}_D < 2 \times 10^6$. The uncertainty in the values of Nusselt number obtained from these relations is ±15 percent. Note that all properties except Pr_s are to be evaluated at the arithmetic mean temperature of the fluid determined from

$$T_m = \frac{T_i + T_e}{2} \tag{7–43}$$

where T_i and T_e are the fluid temperatures at the inlet and the exit of the tube bank, respectively.

The average Nusselt number relations in Table 7–2 are for tube banks with more than 16 rows. Those relations can also be used for tube banks with $N_L < 16$, provided that they are modified as

$$\text{Nu}_{D,\, N_L<16} = F\text{Nu}_D \tag{7–44}$$

where F is a *correction factor* F whose values are given in Table 7–3. For $\text{Re}_D > 1000$, the correction factor is independent of Reynolds number.

Once the Nusselt number and thus the average heat transfer coefficient for the entire tube bank is known, the heat transfer rate can be determined from

TABLE 7–2

Nusselt number correlations for crossflow over tube banks for $N_L > 16$, $0 < \text{Re}_D < 2 \times 10^6$ and $0.7 < \text{Pr} < 500$*

Arrangement	Range of Re_D	Correlation
In-line	0–100	$\text{Nu}_D = 0.9\,\text{Re}_D^{0.4}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	100–1000	$\text{Nu}_D = 0.52\,\text{Re}_D^{0.5}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	$1000–2 \times 10^5$	$\text{Nu}_D = 0.27\,\text{Re}_D^{0.63}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	$2 \times 10^5–2 \times 10^6$	$\text{Nu}_D = 0.333\,\text{Re}_D^{0.8}\,\text{Pr}^{0.4}(\text{Pr}/\text{Pr}_s)^{0.25}$
Staggered	0–500	$\text{Nu}_D = 1.04\,\text{Re}_D^{0.4}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	500–1000	$\text{Nu}_D = 0.71\,\text{Re}_D^{0.5}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	$1000–2 \times 10^5$	$\text{Nu}_D = 0.35(S_T/S_L)^{0.2}\,\text{Re}_D^{0.6}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$
	$2 \times 10^5–2 \times 10^6$	$\text{Nu}_D = 0.031(S_T/S_L)^{0.2}\,\text{Re}_D^{0.8}\,\text{Pr}^{0.36}(\text{Pr}/\text{Pr}_s)^{0.25}$

Source: Zukauskas, 1987.

*All properties except Pr_s are to be evaluated at the arithmetic mean of the inlet and outlet temperatures of the fluid (Pr_s is to be evaluated at T_s).

TABLE 7–3

Correction factor F to be used in $\text{Nu}_{D,\, N_L<16} = F\text{Nu}_D$ for $N_L < 16$ and $\text{Re}_D > 1000$

N_L	1	2	3	4	5	7	10	13
In-line	0.70	0.80	0.86	0.90	0.93	0.96	0.98	0.99
Staggered	0.64	0.76	0.84	0.89	0.93	0.96	0.98	0.99

Source: Zukauskas, 1987.

Newton's law of cooling using a suitable temperature difference ΔT. The first thought that comes to mind is to use $\Delta T = T_s - T_{avg} = T_s - (T_i + T_e)/2$. But this, in general, overpredicts the heat transfer rate. We show in Chap. 8 that the proper temperature difference for internal flow (flow over tube banks is still internal flow through the shell) is the *log mean temperature difference* ΔT_{lm}, defined as

$$\Delta T_{lm} = \frac{(T_s - T_e) - (T_s - T_i)}{\ln[(T_s - T_e)/(T_s - T_i)]} = \frac{\Delta T_e - \Delta T_i}{\ln(\Delta T_e/\Delta T_i)} \tag{7–45}$$

We also show that the exit temperature of the fluid T_e can be determined from

$$T_e = T_s - (T_s - T_i)\exp\left(-\frac{A_s h}{\dot{m}c_p}\right) \tag{7–46}$$

where $A_s = N\pi DL$ is the heat transfer surface area and $\dot{m} = \rho V(N_T S_T L)$ is the mass flow rate of the fluid. Here N is the total number of tubes in the bank, which is the product of N_T (number of tubes in the transverse plane) and N_L (number of rows in the flow direction), L is the length of the tubes, and V is the velocity of the fluid just before entering the tube bank. Then the heat transfer rate can be determined from

$$\dot{Q} = hA_s\Delta T_{lm} = \dot{m}c_p(T_e - T_i) \tag{7–47}$$

The second relation is usually more convenient to use since it does not require the calculation of ΔT_{lm}.

Pressure Drop

Another quantity of interest associated with tube banks is the *pressure drop* ΔP, which is the irreversible pressure loss between the inlet and the exit of the tube bank. It is a measure of the resistance the tubes offer to flow over them and is expressed as

$$\Delta P = N_L f\chi \frac{\rho V_{max}^2}{2} \tag{7–48}$$

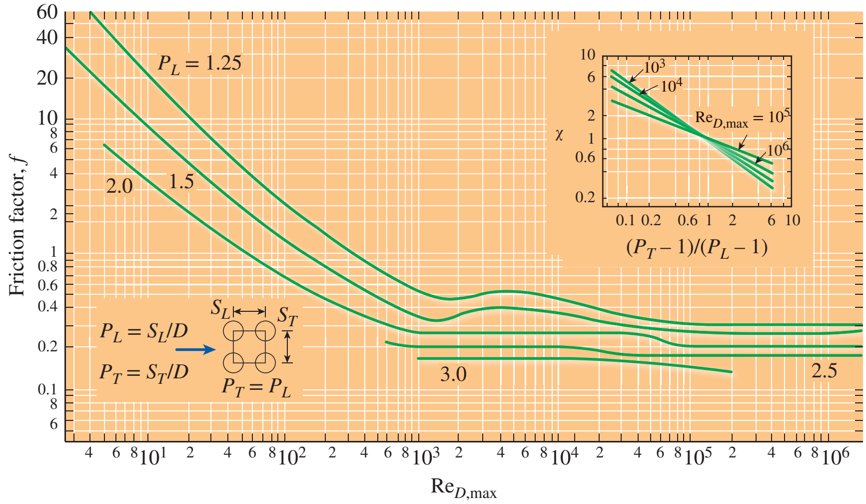
where f is the friction factor and χ is the correction factor, both plotted in Figs. 7–27*a* and 7–27*b* against the Reynolds number based on the maximum velocity V_{max}. The friction factor in Fig. 7–27*a* is for a *square* in-line tube bank ($S_T = S_L$), and the correction factor given in the insert is used to account for the effects of deviation of rectangular in-line arrangements from square arrangement. Similarly, the friction factor in Fig. 7–27*b* is for an *equilateral* staggered tube bank ($S_T = S_D$), and the correction factor is to account for the effects of deviation from equilateral arrangement. Note that $\chi = 1$ for both square and equilateral triangle arrangements. Also, pressure drop occurs in the flow direction, and thus we used N_L (the number of rows) in the ΔP relation.

The power required to move a fluid through a tube bank is proportional to the pressure drop, and when the pressure drop is available, the pumping power required to overcome flow resistance can be determined from

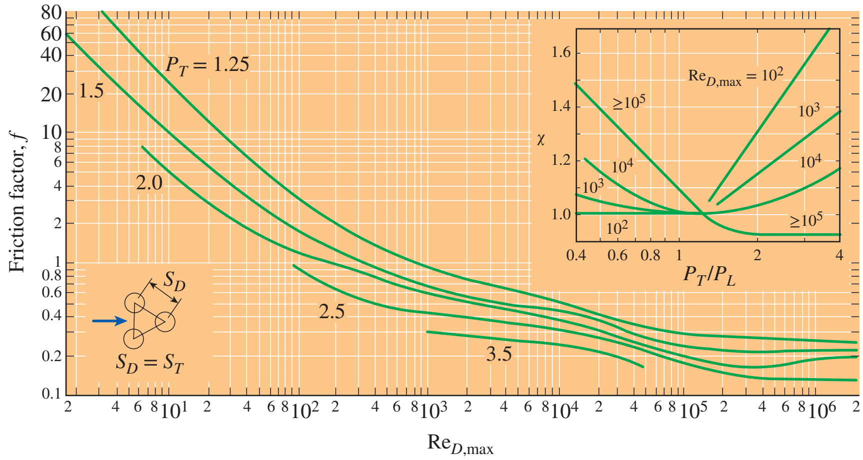
$$\dot{W}_{\text{pump}} = \dot{V}\Delta P = \frac{\dot{m}\Delta P}{\rho} \tag{7–49}$$

where $\dot{V} = V(N_T S_T L)$ is the volume flow rate and $\dot{m} = \rho\dot{V} = \rho V(N_T S_T L)$ is the mass flow rate of the fluid through the tube bank. Note that the power required to keep a fluid flowing through the tube bank (and thus the operating cost) is proportional to the pressure drop. Therefore, the benefits of enhancing heat transfer in a tube bank via rearrangement should be weighed against the cost of additional power requirements.

In this section we limited our consideration to tube banks with base surfaces (no fins). Tube banks with finned surfaces are also commonly used in practice, especially when the fluid is a gas, and heat transfer and pressure drop correlations can be found in the literature for tube banks with pin fins, plate fins, strip fins, etc.



(*a*) In-line arrangement



(*b*) Staggered arrangement

FIGURE 7–27
Friction factor f and correction factor χ for tube banks.
Source: From Zukauskas and Ulinskas (1985).

EXAMPLE 7–8 Preheating Air by Geothermal Water in a Tube Bank

In an industrial facility, air is to be preheated before entering a furnace by geothermal water at 120°C flowing through the tubes of a tube bank located in a duct. Air enters the duct at 20°C and 1 atm with a mean velocity of 4.5 m/s and flows over the tubes in normal direction. The outer diameter of the tubes is 1.5 cm, and the tubes are arranged in-line with longitudinal and transverse pitches of $S_L = S_T = 5$ cm. There are 6 rows in the flow direction with 10 tubes in each row, as shown in Fig. 7–28. Determine the rate of heat transfer per unit length of the tubes and the pressure drop across the tube bank.

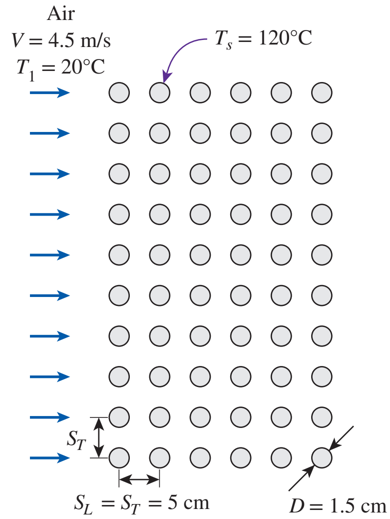


FIGURE 7–28
Schematic for Example 7–8.

SOLUTION Air is heated by geothermal water in a tube bank. The rate of heat transfer to air and the pressure drop of air are to be determined.

Assumptions **1** Steady operating conditions exist. **2** The surface temperature of the tubes is equal to the temperature of geothermal water.

Properties The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of 60°C (will be checked later) and 1 atm (Table A–15):

$$k = 0.02808 \text{ W/m·K} \qquad \rho = 1.059 \text{ kg/m}^3$$

$$c_p = 1.007 \text{ kJ/kg·K} \qquad \text{Pr} = 0.7202$$

$$\mu = 2.008 \times 10^{-5} \text{ kg/m·s} \qquad \text{Pr}_s = \text{Pr}_{@120°\text{C}} = 0.7073$$

Also, the density of air at the inlet temperature of 20°C (for use in the mass flow rate calculation at the inlet) is $\rho_1 = 1.204$ kg/m^3.

Analysis It is given that $D = 0.015$ m, $S_L = S_T = 0.05$ m, and $V = 4.5$ m/s. Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D}V = \frac{0.05}{0.05 - 0.015}(4.5 \text{ m/s}) = 6.43 \text{ m/s}$$

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{(1.059 \text{ kg/m}^3)(6.43 \text{ m/s})(0.015 \text{ m})}{2.008 \times 10^{-5} \text{ kg/m·s}} = 5086$$

The average Nusselt number is determined using the proper relation from Table 7–2 to be

$$\text{Nu}_D = 0.27 \text{ Re}_D^{0.63} \text{ Pr}^{0.36} (\text{Pr/Pr}_s)^{0.25}$$

$$= 0.27 \,(5086)^{0.63}(0.7202)^{0.36}(0.7202/0.7073)^{0.25} = 52.1$$

This Nusselt number is applicable to tube banks with $N_L > 16$. In our case, the number of rows is $N_L = 6$, and the corresponding correction factor from Table 7–3 is $F = 0.945$. Then the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,\, N_L<16} = F\text{Nu}_D = (0.945)(52.1) = 49.3$$

$$h = \frac{\text{Nu}_{D,\, N_L<16}\, k}{D} = \frac{49.3(0.02808\text{ W/m·K})}{0.015\text{ m}} = 92.2\text{ W/m}^2\text{·K}$$

The total number of tubes is $N = N_L \times N_T = 6 \times 10 = 60$. For a unit tube length ($L = 1$ m), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 60\pi(0.015\text{ m})(1\text{ m}) = 2.827\text{ m}^2$$

$$\dot m = \dot m_1 = \rho_1 V(N_T S_T L)$$

$$= (1.204\text{ kg/m}^3)(4.5\text{ m/s})(10)(0.05\text{ m})(1\text{ m}) = 2.709\text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i)\exp\left(-\frac{A_s h}{\dot m c_p}\right)$$

$$= 120 - (120 - 20)\exp\left[-\frac{(2.827\text{ m}^2)(92.2\text{ W/m}^2\text{·K})}{(2.709\text{ kg/s})(1007\text{ J/kg·K})}\right] = 29.11°\text{C}$$

$$\Delta T_{\text{lm}} = \frac{(T_s - T_e) - (T_s - T_i)}{\ln[(T_s - T_e)/(T_s - T_i)]} = \frac{(120 - 29.11) - (120 - 20)}{\ln[(120 - 29.11)/(120 - 20)]} = 95.4°\text{C}$$

$$\dot Q = hA_s\Delta T_{\text{lm}} = (92.2\text{ W/m}^2\text{·K})(2.827\text{ m}^2)(95.4°\text{C}) = \mathbf{2.49 \times 10^4\ W}$$

The rate of heat transfer can also be determined in a simpler way from

$$\dot Q = hA_s\Delta T_{\text{lm}} = \dot m c_p(T_e - T_i)$$

$$= (2.709\text{ kg/s})(1007\text{ J/kg·K})(29.11 - 20)°\text{C} = 2.49 \times 10^4\text{ W}$$

For this square in-line tube bank, the friction coefficient corresponding to $\text{Re}_D = 5086$ and $S_L/D = 5/1.5 = 3.33$ is, from Fig. 7–27*a*, $f = 0.16$. Also, $\chi = 1$ for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f\chi \frac{\rho V_{\max}^2}{2}$$

$$= 6(0.16)(1)\frac{(1.059\text{ kg/m}^3)(6.43\text{ m/s})^2}{2}\left(\frac{1\text{ N}}{1\text{ kg·m/s}^2}\right) = \mathbf{21\ Pa}$$

Discussion The arithmetic mean fluid temperature is $(T_i + T_e)/2 = (20 + 29.11)/2 = 24.6°\text{C}$, which is not close to the assumed value of 60°C. Repeating calculations for 25°C gives 2.57×10^4 W for the rate of heat transfer and 23.5 Pa for the pressure drop.

SUMMARY

The force a flowing fluid exerts on a body in the flow direction is called *drag*. The part of drag that is due directly to wall shear stress τ_w is called the *skin friction drag* since it is caused by frictional effects, and the part that is due directly to pressure is called the *pressure drag* or *form drag* because of its strong dependence on the form or shape of the body.

The *drag coefficient* C_D is a dimensionless number that represents the drag characteristics of a body, and it is defined as

$$C_D = \frac{F_D}{\frac{1}{2}\rho V^2 A}$$

where A is the *frontal area* for blunt bodies and the surface area for parallel flow over flat plates or thin airfoils. For flow over a flat plate, the Reynolds number is

$$\text{Re}_x = \frac{\rho V x}{\mu} = \frac{Vx}{\nu}$$

Transition from laminar to turbulent occurs at the *critical Reynolds number* of

$$\text{Re}_{x,\,\text{cr}} = \frac{\rho V x_{\text{cr}}}{\mu} = 5 \times 10^5$$

For parallel flow over an isothermal flat plate, the local friction coefficent and Nusselt number are

Laminar: $$C_{f,x} = \frac{0.664}{\text{Re}_x^{1/2}} \qquad \text{Re}_x < 5 \times 10^5$$

$$\text{Nu}_x = \frac{h_x x}{k} = 0.332\text{Re}_x^{0.5}\,\text{Pr}^{1/3} \qquad \text{Pr} > 0.6$$

Turbulent:

$$C_{f,x} = \frac{0.059}{\text{Re}_x^{1/5}} \qquad 5 \times 10^5 \le \text{Re}_x \le 10^7$$

$$\text{Nu}_x = \frac{h_x x}{k} = 0.0296\,\text{Re}_x^{0.8}\,\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5 \times 10^5 \le \text{Re}_x \le 10^7 \end{array}$$

The *average* friction coefficient relations for flow over a flat plate are:

Laminar: $$C_f = \frac{1.33}{\text{Re}_L^{1/2}} \qquad \text{Re}_L < 5 \times 10^5$$

Turbulent: $$C_f = \frac{0.074}{\text{Re}_L^{1/5}} \qquad 5 \times 10^5 \le \text{Re}_L \le 10^7$$

Combined: $$C_f = \frac{0.074}{\text{Re}_L^{1/5}} - \frac{1742}{\text{Re}_L} \qquad 5 \times 10^5 \le \text{Re}_L \le 10^7$$

Rough surface, turbulent: $$C_f = \left(1.89 - 1.62\log\frac{\varepsilon}{L}\right)^{-2.5}$$

The average Nusselt number relations for flow over an isothermal flat plate are:

Laminar: $$\text{Nu} = \frac{hL}{k} = 0.664\,\text{Re}_L^{0.5}\,\text{Pr}^{1/3} \qquad \begin{array}{l} \text{Pr} > 0.6 \\ \text{Re}_L < 5 \times 10^5 \end{array}$$

Turbulent:

$$\text{Nu} = \frac{hL}{k} = 0.037\,\text{Re}_L^{0.8}\,\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5 \times 10^5 \le \text{Re}_L \le 10^7 \end{array}$$

Combined:

$$\text{Nu} = \frac{hL}{k} = (0.037\,\text{Re}_L^{0.8} - 871)\,\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5 \times 10^5 \le \text{Re}_L \le 10^7 \end{array}$$

For isothermal surfaces with an unheated starting section of length ξ, the local and average Nusselt number relations are

Laminar: $$\text{Nu}_x = \frac{\text{Nu}_{x\,(\text{for }\xi=0)}}{[1-(\xi/x)^{3/4}]^{1/3}} = \frac{0.332\,\text{Re}_x^{0.5}\,\text{Pr}^{1/3}}{[1-(\xi/x)^{3/4}]^{1/3}}$$

Turbulent: $$\text{Nu}_x = \frac{\text{Nu}_{x\,(\text{for }\xi=0)}}{[1-(\xi/x)^{9/10}]^{1/9}} = \frac{0.0296\,\text{Re}_x^{0.8}\,\text{Pr}^{1/3}}{[1-(\xi/x)^{9/10}]^{1/9}}$$

Laminar: $$\text{Nu} = \text{Nu}_{(\text{for }\xi=0)}\frac{L}{L-\xi}[1-(\xi/L)^{3/4}]^{2/3}$$

Turbulent: $$\text{Nu} = \text{Nu}_{(\text{for }\xi=0)}\frac{L}{L-\xi}[1-(\xi/L)^{9/10}]^{8/9}$$

These relations are for the case of *isothermal* surfaces. When a flat plate is subjected to *uniform heat flux,* the local Nusselt number is given by

Laminar: $$\text{Nu}_x = 0.453\,\text{Re}_x^{0.5}\,\text{Pr}^{1/3} \qquad \begin{array}{l} \text{Pr} > 0.6 \\ \text{Re}_x < 5 \times 10^5 \end{array}$$

Turbulent: $$\text{Nu}_x = 0.0308\,\text{Re}_x^{0.8}\,\text{Pr}^{1/3} \qquad \begin{array}{l} 0.6 \le \text{Pr} \le 60 \\ 5 \times 10^5 \le \text{Re}_x \le 10^7 \end{array}$$

The average Nusselt numbers for crossflow over a *cylinder* and *sphere* are

$$\text{Nu}_{\text{cyl}} = \frac{hD}{k} = 0.3 + \frac{0.62\,\text{Re}^{1/2}\,\text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}}\left[1 + \left(\frac{\text{Re}}{282{,}000}\right)^{5/8}\right]^{4/5}$$

which is valid for Re Pr > 0.2, and

$$\text{Nu}_{\text{sph}} = \frac{hD}{k} = 2 + [0.4\,\text{Re}^{1/2} + 0.06\,\text{Re}^{2/3}]\text{Pr}^{0.4}\left(\frac{\mu_\infty}{\mu_s}\right)^{1/4}$$

which is valid for $3.5 \le \text{Re} \le 8 \times 10^4$, $0.7 \le \text{Pr} \le 380$ and $1.0 \le (\mu_\infty/\mu_s) \le 3.2$. The fluid properties are evaluated at the film temperature $T_f = (T_\infty + T_s)/2$ in the case of a cylinder, and at the free-stream temperature T_∞ (except for μ_s, which is evaluated at the surface temperature T_s) in the case of a sphere.

In tube banks, the Reynolds number is based on the maximum velocity $V_{\max}$ that is related to the approach velocity V as

In-line and *Staggered* with $S_D < (S_T + D)/2$:

$$V_{\max} = \frac{S_T}{S_T - D}V$$

Staggered with $S_D < (S_T + D)/2$:

$$V_{\max} = \frac{S_T}{2(S_D - D)}V$$

where S_T is the transverse pitch and S_D is the diagonal pitch. The average Nusselt number for crossflow over tube banks is expressed as

$$\text{Nu}_D = \frac{hD}{k} = C\,\text{Re}_D^m\,\text{Pr}^n\,(\text{Pr}/\text{Pr}_s)^{0.25}$$

where the values of the constants C, m, and n depend on Reynolds number. Such correlations are given in Table 7–2. All properties except Pr_s are to be evaluated at the arithmetic mean of the inlet and exit temperatures of the fluid defined as $T_m = (T_i + T_e)/2$.

The average Nusselt number for tube banks with less than 16 rows is expressed as

$$\text{Nu}_{D,\,N_L<16} = F\text{Nu}_D$$

where F is the *correction factor* whose values are given in Table 7–3. The heat transfer rate to or from a tube bank is determined from

$$\dot{Q} = hA_s\Delta T_{\text{lm}} = \dot{m}c_p(T_e - T_i)$$

where ΔT_{lm} is the logarithmic mean temperature difference defined as

$$\Delta T_{\text{lm}} = \frac{(T_s - T_e) - (T_s - T_i)}{\ln[(T_s - T_e)/(T_s - T_i)]} = \frac{\Delta T_e - \Delta T_i}{\ln(\Delta T_e/\Delta T_i)}$$

and the exit temperature of the fluid T_e is

$$T_e = T_s - (T_s - T_i)\exp\left(-\frac{A_s h}{\dot{m}c_p}\right)$$

where $A_s = N\pi DL$ is the heat transfer surface area and $\dot{m} = \rho V(N_T S_T L)$ is the mass flow rate of the fluid. The pressure drop ΔP for a tube bank is expressed as

$$\Delta P = N_L f\chi\frac{\rho V_{\max}^2}{2}$$

where f is the friction factor and χ is the correction factor, both given in Fig. 7–27.

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PROBLEMS*

Drag Force and Heat Transfer in External Flow

7–1C What is the difference between the upstream velocity and the free-stream velocity? For what types of flow are these two velocities equal to each other?

7–2C What is drag? What causes it? Why do we usually try to minimize it?

7–3C What is lift? What causes it? Does wall shear contribute to the lift?

7–4C During flow over a given body, the drag force, the upstream velocity, and the fluid density are measured. Explain how you would determine the drag coefficient. What area would you use in calculations?

7–5C Define the frontal area of a body subjected to external flow. When is it appropriate to use the frontal area in drag and lift calculations?

7–6C What is the difference between skin friction drag and pressure drag? Which is usually more significant for slender bodies such as airfoils?

7–7C What is the difference between streamlined and blunt bodies? Is a tennis ball a streamlined or blunt body?

7–8C What is the effect of streamlining on (*a*) friction drag and (*b*) pressure drag? Does the total drag acting on a body necessarily decrease as a result of streamlining? Explain.

7–9C What is the effect of surface roughness on the friction drag coefficient in laminar and turbulent flows?

7–10C What is flow separation? What causes it? What is the effect of flow separation on the drag coefficient?

* Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the PtD icon are Prevention through Design problems. Problems with the C&S icon are Codes and Standards problems.

Flow Over Flat Plates

7–11C What does the friction coefficient represent in flow over a flat plate? How is it related to the drag force acting on the plate?

7–12C Consider laminar flow over a flat plate. Will the friction coefficient change with distance from the leading edge? How about the heat transfer coefficient?

7–13C How are the average friction and heat transfer coefficients determined in flow over a flat plate?

7–14 Air at 25°C and 1 atm is flowing over a long flat plate with a velocity of 8 m/s. Determine the distance from the leading edge of the plate where the flow becomes turbulent and the thickness of the boundary layer at that location.

7–15 Repeat Prob. 7–14 for water.

7–16 The weight of a thin flat plate 50-cm × 50-cm is balanced by a counterweight that has a mass of 2 kg, as shown in the figure. Now a fan is turned on, and air at 1 atm and 25°C flows downward over both surfaces of the plate with a free-stream velocity of 8 m/s. Determine the mass of the counterweight that needs to be added in order to balance the plate in this case.

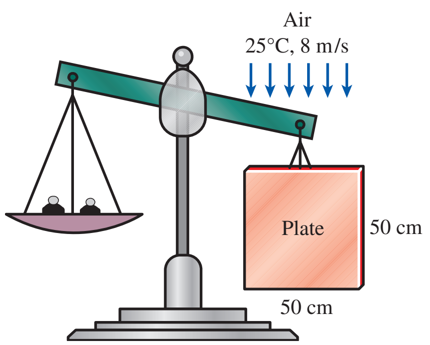

FIGURE P7–16

7–17 Air at 15°C and 1 atm flows over a 0.3-m-wide plate at 65°C at a velocity of 3.0 m/s. Compute the following quantities at $x = 0.3$ m:

(*a*) Hydrodynamic boundary layer thickness, m
(*b*) Local friction coefficient
(*c*) Average friction coefficient
(*d*) Total drag force due to friction, N
(*e*) Local convection heat transfer coefficient, W/m²·K
(*f*) Average convection heat transfer coefficient, W/m²·K
(*g*) Rate of convective heat transfer, W

7–18 Engine oil at 85°C flows over a 10-m-long flat plate whose temperature is 35°C with a velocity of 2.5 m/s. Determine the total drag force and the rate of heat transfer over the entire plate per unit width.

7–19E Air at 60°F flows over a 10-ft-long flat plate at 7 ft/s. Determine the local friction and heat transfer coefficients at intervals of 1 ft, and plot the results against the distance from the leading edge.

7–20E Reconsider Prob. 7–19E. Using appropriate software, evaluate the local friction and heat transfer coefficients along the plate at intervals of 0.1 ft, and plot them against the distance from the leading edge.

7–21 Consider laminar flow of a fluid over a flat plate maintained at a constant temperature. Now the free-stream velocity of the fluid is doubled. Determine the change in the drag force on the plate and the rate of heat transfer between the fluid and the plate. Assume the flow to remain laminar.

7–22 In an experiment, the local heat transfer over a flat plate was correlated in the form of the local Nusselt number as expressed by the following correlation

$$\text{Nu}_x = 0.035\text{Re}_x^{0.8}\text{Pr}^{1/3}$$

Determine the ratio of the average convection heat transfer coefficient (h) over the entire plate length to the local convection heat transfer coefficient (h_x) at $x = L$.

7–23 C&S Hot water vapor flows in parallel over the upper surface of a 1-m-long plate. The velocity of the water vapor is 10 m/s at a temperature of 450°C. A copper–silicon (ASTM B98) bolt is embedded in the plate at mid-length. The maximum use temperature for the ASTM B98 copper–silicon bolt is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). To devise a cooling mechanism to keep the bolt from getting above the maximum use temperature, it becomes necessary to determine the local heat flux at the location where the bolt is embedded. If the plate surface is kept at the maximum use temperature of the bolt, what is the local heat flux from the hot water vapor at the location of the bolt?

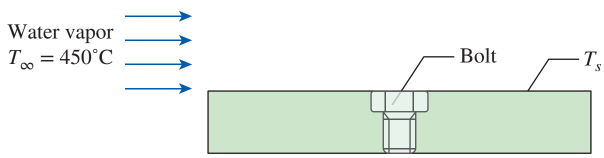

FIGURE P7–23

7–24 Water at 43.3°C flows over a large plate at a velocity of 30.0 cm/s. The plate is 1.0-m long (in the flow direction), and its surface is maintained at a uniform temperature of 10.0°C. Calculate the steady rate of heat transfer per unit width of the plate.

7–25 The forming section of a plastics plant puts out a continuous sheet of plastic that is 1.2-m wide and 2-mm thick at a rate of 15 m/min. The temperature of the plastic sheet is

90°C when it is exposed to the surrounding air, and the sheet is subjected to airflow at 30°C at a velocity of 3 m/s on both sides along its surfaces normal to the direction of motion of the sheet. The width of the air cooling section is such that a fixed point on the plastic sheet passes through that section in 2 s. Determine the rate of heat transfer from the plastic sheet to the air.

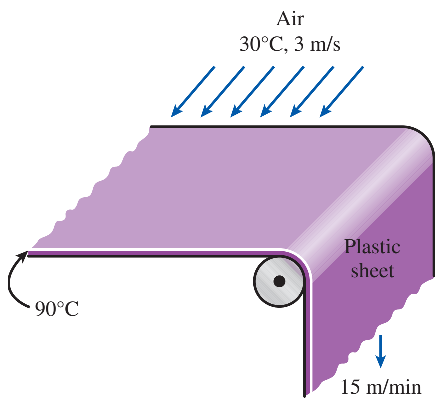

FIGURE P7–25

7–26 Hot carbon dioxide exhaust gas at 1 atm is being cooled by flat plates. The gas at 220°C flows in parallel over the upper and lower surfaces of a 1.5-m-long flat plate at a velocity of 3 m/s. If the flat plate surface temperature is maintained at 80°C, determine (*a*) the local convection heat transfer coefficient at 1 m from the leading edge, (*b*) the average convection heat transfer coefficient over the entire plate, and (*c*) the total heat flux transfer to the plate.

7–27 C&S The upper surface of a 1-m × 1-m ASTM B152 copper plate is being cooled by air at 20°C. The air is flowing parallel over the plate surface at a velocity of 0.5 m/s. The maximum use temperature for the ASTM B152 copper plate is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Determine the average convection heat transfer rate from the plate surface that is required to keep the plate surface temperature from going higher than the maximum use temperature.

7–28 A transformer that is 10-cm long, 6.2-cm wide, and 5-cm high is to be cooled by attaching a 10-cm × 6.2-cm-wide polished aluminum heat sink (emissivity = 0.03) to its top surface. The heat sink has seven fins, which are 5-mm high, 2-mm thick, and 10-cm long. A fan blows air at 25°C parallel to the passages between the fins. The heat sink is to dissipate 12 W of heat, and the base temperature of the heat sink is not to exceed 60°C. Assuming the fins and the base plate to be nearly isothermal and the radiation heat transfer to be negligible, determine the minimum free-stream velocity the fan needs to supply to avoid overheating. Assume the flow is laminar over the entire finned surface of the transformer.

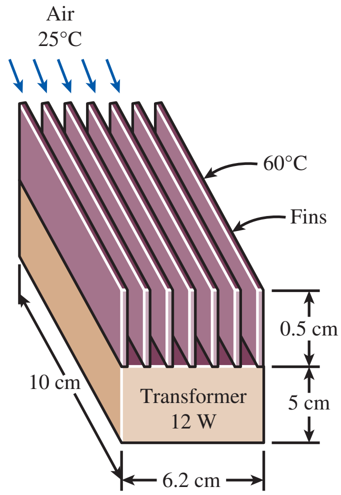

FIGURE P7–28

7–29 Repeat Prob. 7–28 assuming the heat sink to be black-anodized and thus to have an effective emissivity of 0.90. Note that in radiation calculations the base area (10-cm × 6.2-cm) is to be used, not the total surface area.

7–30 Hot engine oil at 150°C is flowing in parallel over a flat plate at a velocity of 2 m/s. Surface temperature of the 0.5-m-long flat plate is constant at 50°C. Determine (*a*) the local convection heat transfer coefficient at 0.2 m from the leading edge and the average convection heat transfer coefficient, and (*b*) repeat part (*a*) using the Churchill and Ozoe (1973) relation.

7–31 Parallel plates form a solar collector that covers a roof, as shown in the figure. The plates are maintained at 15°C, while ambient air at 10°C flows over the roof with $V = 2$ m/s. Determine the rate of convective heat loss from (*a*) the first plate and (*b*) the third plate.

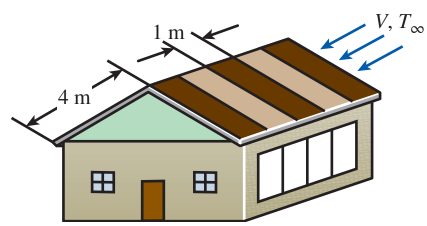

FIGURE P7–31

7–32 Hydrogen gas at 1 atm is flowing in parallel over the upper and lower surfaces of a 3-m-long flat plate at a velocity of 2.5 m/s. The gas temperature is 120°C, and the surface temperature of the plate is maintained at 30°C. Using appropriate software, investigate the local convection heat transfer coefficient and the local total convection heat flux along the plate. By varying the location along the plate for $0.2 \leq x \leq 3$ m, plot the local convection heat transfer coefficient and the local total convection heat flux as functions of x. Assume flow is laminar, but make sure to verify this assumption.

7–33 Carbon dioxide and hydrogen as ideal gases at 1 atm and −20°C flow in parallel over a flat plate. The flow velocity of each gas is 1 m/s, and the surface temperature of the 3-m-long plate is maintained at 20°C. Using appropriate software, evaluate the local Reynolds number, the local Nusselt number, and the local convection heat transfer coefficient along the plate for each gas. By varying the location along the plate for $0.2 \leq x \leq 3$ m, plot the local Reynolds number, the local Nusselt number, and the local convection heat transfer coefficient for each gas as functions of x. Discuss which gas has a higher local Nusselt number and which gas has a higher convection heat transfer coefficient along the plate. Assume flow is laminar, but make sure to verify this assumption.

7–34 An array of power transistors, dissipating 5 W of power each, are to be cooled by mounting them on a 25-cm × 25-cm-square aluminum plate and blowing air at 35°C over the plate with a fan at a velocity of 4 m/s. The average temperature of the plate is not to exceed 65°C. Assuming the heat transfer from the back side of the plate to be negligible and disregarding radiation, determine the number of transistors that can be placed on this plate.

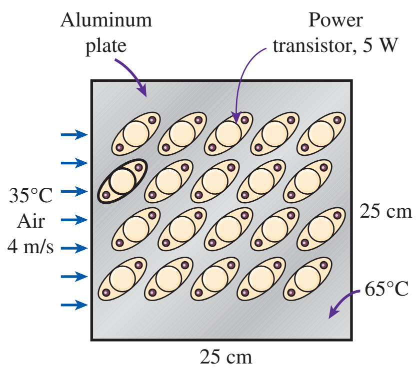

FIGURE P7–34

7–35 Repeat Prob. 7–34 for a location at an elevation of 1610 m where the atmospheric pressure is 83.4 kPa. *Answer:* 5

7–36 C&S Air flows in parallel over the upper surface of a 2-m-long plate. The velocity of the air is 20 m/s at a temperature of 250°C. Two copper–silicon (ASTM B98) bolts are embedded in the plate: the first bolt at 0.25 m from the leading edge, and the second bolt at 1.75 m from the leading edge. The maximum use temperature for the ASTM B98 copper–silicon bolt is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). To devise a cooling mechanism to keep the bolts from heating above the maximum use temperature, it becomes necessary to determine the local heat fluxes at the locations where the bolts are embedded. If the plate surface is kept at the maximum use temperature of the bolts, what are the local heat fluxes from the air at the locations of both bolts?

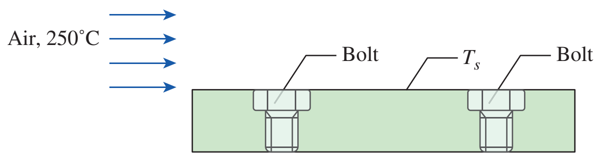

FIGURE P7–36

7–37E Consider a refrigeration truck traveling at 70 mph at a location where the air temperature is 80°F. The refrigerated compartment of the truck can be considered to be a 9-ft-wide, 7-ft-high, and 20-ft-long rectangular box. The refrigeration system of the truck can provide 3 tons of refrigeration (i.e., it can remove heat at a rate of 600 Btu/min). The outer surface of the truck is coated with a low-emissivity material, and thus radiation heat transfer is very small. Determine the average temperature of the outer surface of the refrigeration compartment of the truck if the refrigeration system is observed to be operating at half the capacity. Assume the airflow over the entire outer surface to be turbulent and the heat transfer coefficient at the front and rear surfaces to be equal to that on side surfaces. For air properties evaluations, assume a film temperature of 80°F. Is this a good assumption?

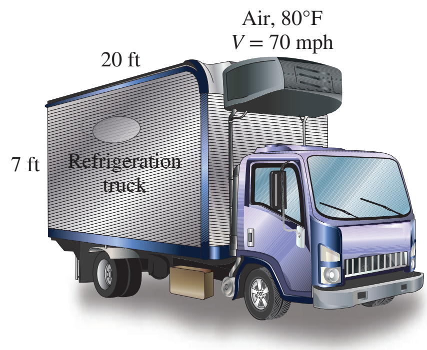

FIGURE P7–37E

7–38 Consider a hot automotive engine, which can be approximated as a 0.5-m-high, 0.40-m-wide, and 0.8-m-long rectangular block. The bottom surface of the block is at a temperature of 100°C and has an emissivity of 0.95. The ambient air is at 20°C, and the road surface is at 25°C. Determine the rate of heat transfer from the bottom surface of the engine block by convection and radiation as the car travels at a velocity of 80 km/h. Assume the flow to be turbulent over the entire surface because of the constant agitation of the engine block.

7–39 Air at 1 atm is flowing in parallel over a 3-m-long flat plate with a velocity of 7 m/s. The air has a free-stream temperature of 120°C, and the surface temperature of the plate is maintained at 20°C. Determine the distance x from the leading edge of the plate where the critical Reynolds number ($Re_{cr} = 5 \times 10^5$) is reached. Then, using appropriate software, evaluate the local convection heat transfer coefficient along the plate. By varying the location along the plate for $0.2 \leq x \leq 3$ m, plot the local convection heat transfer coefficient as a function of x, and discuss the results.

7–40 Heat dissipated from a machine in operation can cause hot spots on its surface. Exposed hot spots that can cause thermal burns on human skin are considered to be hazards in the workplace. Consider a 1.5-m × 1.5-m flat machine surface that is made of 5-mm-thick aluminum ($k = 237$ W/m·K). During operation, the machine's inner aluminum surface temperature can be as high as 90°C, while the outer surface is cooled with 30°C air flowing in parallel over it at 10 m/s. To protect machine operators from thermal burns, the machine surface can be covered with insulation. The aluminum/insulation interface has a thermal contact conductance of 3000 W/m²·K. What is the thickness of insulation (with a thermal conductivity of 0.06 W/m·K) needed to keep the local outer surface temperature at 45°C or lower? Using appropriate software, plot the required insulation thickness and the local convection heat transfer coefficient along the outer surface in parallel with the airflow.

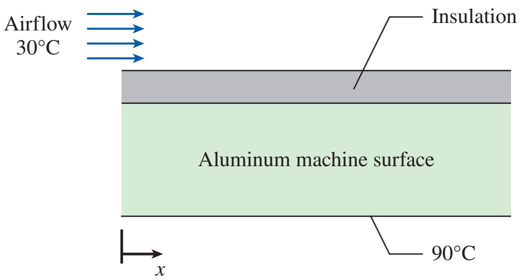

FIGURE P7–40

7–41 Air at 1 atm and 20°C is flowing over the top surface of a 0.5-m-long thin flat plate. The airstream velocity is 50 m/s, and the plate is maintained at a constant surface temperature of 180°C. Determine (*a*) the average friction coefficient and (*b*) the average convection heat transfer coefficient, and (*c*) repeat part (*b*) using the modified Reynolds analogy.

FIGURE P7–41

7–42 C&S In cryogenic equipment, cold air flows in parallel over the surface of a 2-m × 2-m ASTM A240 410S stainless steel plate. The air velocity is 5 m/s at a temperature of −70°C. The minimum temperature suitable for the ASTM A240 410S plate is −30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The plate is heated to keep its surface temperature from going below −30°C. Determine the average heat transfer rate required to keep the plate surface from getting below the minimum suitable temperature.

7–43 A 5-m-long strip of sheet metal is being transported on a conveyor at a velocity of 5 m/s, while the coating on the upper surface is being cured by infrared lamps. The coating on the upper surface of the metal strip has an absorptivity of 0.6 and an emissivity of 0.7, while the surrounding ambient air temperature is 25°C. Radiation heat transfer occurs only on the upper surface, while convection heat transfer occurs on both upper and lower surfaces of the sheet metal. If the infrared lamps supply a constant heat flux of 5000 W/m², determine the surface temperature of the sheet metal. Evaluate the properties of air at 80°C.

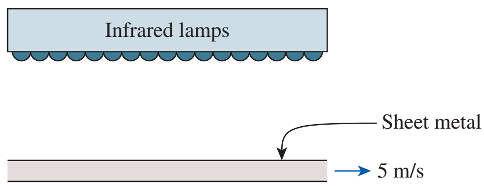

FIGURE P7–43

7–44 Reconsider Prob. 7–43. Using appropriate software, evaluate the effect of the sheet metal velocity on its surface temperature. By varying the sheet metal velocity from 3 to 30 m/s, plot the surface temperature of the sheet metal as a function of velocity. If the surface temperature of the sheet metal should be maintained above 100°C, determine the minimum velocity that is necessary to satisfy this condition. Evaluate the properties of air at 80°C.

7–45 The local atmospheric pressure in Denver, Colorado (elevation 1610 m), is 83.4 kPa. Air at this pressure and at 30°C flows with a velocity of 6 m/s over a 2.5-m × 8-m flat plate whose temperature is 120°C. Determine the rate of heat transfer from the plate if the air flows parallel to the (*a*) 8-m-long side and (*b*) the 2.5-m side.

7–46 During a cold winter day, wind at 42 km/h is blowing parallel to a 6-m-high and 10-m-long wall of a house. If the air outside is at 5°C and the surface temperature of the wall is 12°C, determine the rate of heat loss from that wall by convection. What would your answer be if the wind velocity was doubled? *Answers:* 10.8 kW, 19.4 kW

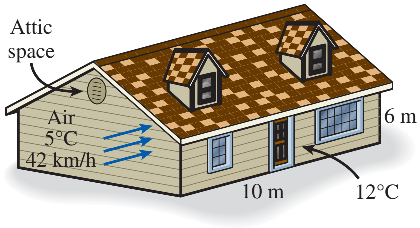

FIGURE P7–46

7–47 Reconsider Prob. 7–46. Using appropriate software, investigate the effects of wind velocity and outside air temperature on the rate of heat loss from the wall by convection. Let the wind velocity vary from 10 km/h to 80 km/h and the outside air temperature from 0°C to 10°C. Plot the rate of heat loss as a function of the wind velocity and of the outside temperature, and discuss the results.

7–48E Warm air is blown over the inner surface of an automobile windshield to defrost ice accumulated on the outer surface of the windshield. Consider an automobile windshield ($k_w = 0.8$ Btu/h·ft·R) with an overall height of 20 in and thickness of 0.2 in. The outside air (1 atm) ambient temperature is 8°F, and the average airflow velocity over the outer windshield surface is 50 mph, while the ambient temperature inside the automobile is 77°F. Determine the value of the convection heat transfer coefficient for the warm air blowing over the inner surface of the windshield that is needed to cause the accumulated ice to begin melting. Assume the windshield surface can be treated as a flat-plate surface.

7–49 The top surface of the passenger car of a train moving at a velocity of 95 km/h is 2.8-m wide and 8-m long. The top surface is absorbing solar radiation at a rate of 380 W/m^2, and the temperature of the ambient air is 30°C. Assuming the roof of the car to be perfectly insulated and the radiation heat exchange with the surroundings to be small relative to convection, determine the equilibrium temperature of the top surface of the car. *Answer:* 37.5°C

FIGURE P7–49

7–50 Reconsider Prob. 7–49. Using appropriate software, investigate the effects of the train velocity and the rate of absorption of solar radiation on the equilibrium temperature of the top surface of the car. Let the train velocity vary from 10 km/h to 120 km/h and the rate of solar absorption from 100 W/m^2 to 500 W/m^2. Plot the equilibrium temperature as functions of train velocity and solar radiation absorption rate, and discuss the results.

7–51 Solar radiation is incident on the glass cover of a solar collector at a rate of 700 W/m^2. The glass transmits 88 percent of the incident radiation and has an emissivity of 0.90. The entire hot water needs of a family in summer can be met by two collectors 1.2-m high and 1-m wide. The two collectors are attached to each other on one side so that they appear like a single collector 1.2-m × 2-m in size. The temperature of the glass cover is measured to be 35°C on a day when the surrounding air temperature is 25°C and the wind is blowing at 30 km/h. The effective sky temperature for radiation exchange between the glass cover and the open sky is −40°C. Water enters the tubes attached to the absorber plate at a rate of 1 kg/min. Assuming the back surface of the absorber plate to be heavily insulated and the only heat loss to occur through the glass cover, determine (*a*) the total rate of heat loss from the collector, (*b*) the collector efficiency, which is the ratio of the amount of heat transferred to the water to the solar energy incident on the collector, and (*c*) the temperature rise of water as it flows through the collector.

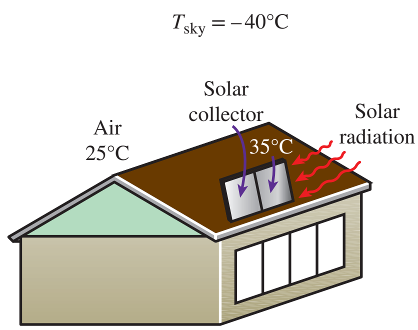

FIGURE P7–51

7–52 PtD Reconsider Example 7–2. Another method that could be explored to keep the engine outer surface temperature below the fire hazard limit of 180°C is to increase the insulation thickness. Increase the insulation thickness by threefold and verify if this method would keep the temperature below the required limit for the safe operation of the engine. Compare the effectiveness of this method with the one used in Example 7–2.

7–53 PtD The outer surface of an engine is situated in a place where oil leakage can occur. When leaked oil comes in contact with a hot surface that has a temperature above its autoignition temperature, the oil can ignite spontaneously. Consider an engine cover that is made of a stainless steel plate with a thickness of 1 cm and a thermal conductivity of 14 W/m·K. The inner surface of the engine cover is exposed to hot air with a convection heat transfer coefficient of 7 W/m^2·K at a temperature of 333°C. The engine outer surface is cooled by air blowing in parallel over the 2-m-long surface at 7.1 m/s, in an environment where the ambient air is at 60°C. To prevent fire hazard in the event of an oil leak on the engine cover, a layer of thermal barrier coating (TBC) with a thermal conductivity of 1.1 W/m·K is applied on the engine cover outer surface. Would a TBC layer with a thickness of 4 mm in conjunction with 7.1 m/s air cooling be sufficient to keep the engine cover surface from going above 180°C to prevent fire hazard? Evaluate the air properties at 120°C.

7–54 PtD Metal plates ($k = 180$ W/m·K, $\rho = 2800$ kg/m^3, and $c_p = 880$ J/kg·K) with a length of 1 m and a thickness of 2 cm exiting an oven are then conveyed through a 10-m-long cooling chamber at a speed of 5 mm/s. The plates enter the cooling chamber at an initial temperature of 155°C. In the cooling chamber, the plates are cooled with 10°C air blowing in parallel over them. To prevent thermal burns, it is necessary to design the cooling process so that the plates exit the cooling chamber at a relatively safe temperature. Determine the air velocity such that the temperature of the plates exiting the cooling chamber is 45°C or less. Assume combined laminar and turbulent flow (verify this assumption). *Hint:* Use lumped system analysis to determine the required cooling time (verify the application of this method to this problem).

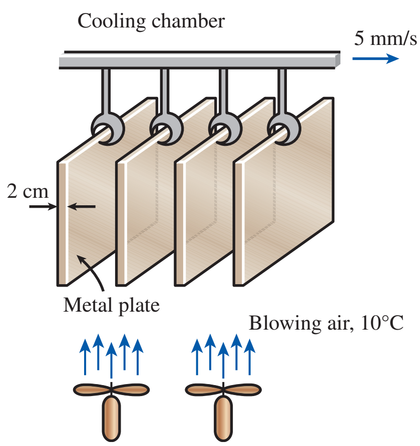

FIGURE P7–54

7–55 Mercury at 25°C flows over a 3-m-long and 2-m-wide flat plate maintained at 75°C with a velocity of 0.01 m/s. Determine the rate of heat transfer from the entire plate.

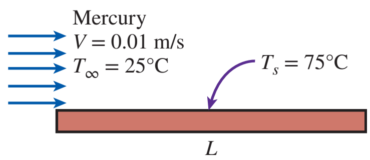

FIGURE P7–55

7–56 Liquid mercury at 250°C is flowing in parallel over a flat plate at a velocity of 0.3 m/s. Surface temperature of the 0.1-m-long flat plate is constant at 50°C. Determine (*a*) the local convection heat transfer coefficient at 5 cm from the leading edge and (*b*) the average convection heat transfer coefficient over the entire plate.

7–57 Water vapor at 250°C is flowing with a velocity of 5 m/s in parallel over a 2-m-long flat plate where there is an unheated starting length of 0.5 m. The heated section of the flat plate is maintained at a constant temperature of 50°C. Determine (*a*) the local convection heat transfer coefficient at the trailing edge, (*b*) the average convection heat transfer coefficient for the heated section, and (*c*) the rate of heat transfer per unit width for the heated section.

7–58 A 15-mm × 15-mm silicon chip is mounted such that the edges are flush in a substrate. The chip dissipates 1.4 W of power uniformly, while air at 20°C (1 atm) with a velocity of 25 m/s is used to cool the upper surface of the chip. If the substrate provides an unheated starting length of 15 mm, determine the surface temperature at the trailing edge of the chip. Evaluate the air properties at 50°C.

7–59 C & S In cryogenic equipment, cold gas flows in parallel over the surface of a 1-m-long plate. The gas velocity is 4 m/s at a temperature of −70°C. The gas has a thermal conductivity of 0.01979 W/m·K, a kinematic viscosity of 9.319×10^{-6} m^2/s, and a Prandtl number of 0.7440. A stainless steel (ASTM A437 B4B) bolt is embedded in the plate at midlength. The minimum temperature suitable for the ASTM A437 B4B stainless steel bolt is −30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). To keep the bolt from getting below its minimum suitable temperature, the plate is subjected to a uniform heat flux of 250 W/m^2. Determine whether the heat flux to the plate is sufficient to keep the bolt above the minimum suitable temperature of −30°C.

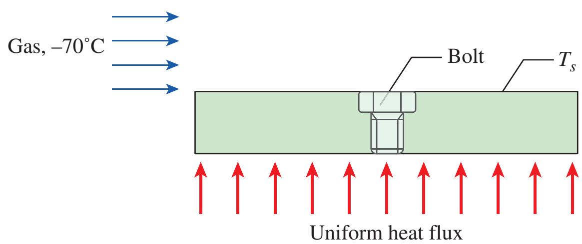

FIGURE P7–59

7–60 The upper surface of a metal plate is being cooled with parallel airflow while its lower surface is subjected to a uniform heat flux of 810 W/m^2. The air has a free-stream velocity and temperature of 2.5 m/s and 15°C, respectively. Determine the surface temperature of the plate at $x = 1.5$ m from the leading edge. *Hint:* The surface temperature has to be found iteratively. Start the iteration process with an initial guess of 45°C for the surface temperature.

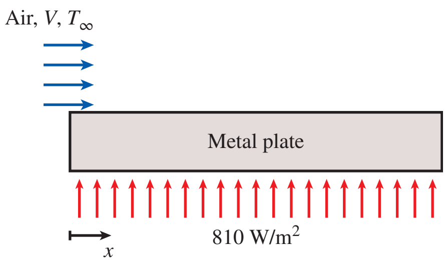

FIGURE P7–60

7–61 Reconsider Prob. 7–60. Using appropriate software, evaluate the local convection heat transfer coefficient, the local surface temperature, and the local film temperature along the plate. By varying the location along the plate for $0.2 \le x \le 3$ m, plot the local convection heat transfer coefficient, the local surface temperature, and the local film temperature as functions of x.

7–62 Air is flowing in parallel over the upper surface of a flat plate with a length of 4 m. The first half of the plate length, from the leading edge, has a constant surface temperature of 50°C. The second half of the plate length is subjected to a uniform heat flux of 86 W/m^2. The air has a free-stream velocity and temperature of 2 m/s and 10°C, respectively. Determine the local convection heat transfer coefficients at 1 m and 3 m from the leading edge. As a first approximation, assume the boundary layer over the second portion of the plate with uniform heat flux has not been affected by the first half of the plate with constant surface temperature. Evaluate the air properties at a film temperature of 30°C. Is the film temperature $T_f = 30$°C applicable at $x = 3$ m?

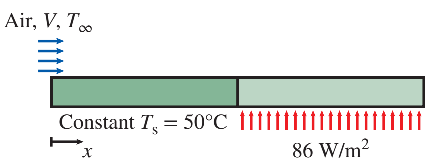

FIGURE P7–62

7–63 Reconsider Prob. 7–62. Using appropriate software, evaluate the local convection heat transfer coefficient, the local surface temperature, and the local film temperature along the plate. By varying the location along the plate for $0.2 \le x \le 4$ m, plot the local convection heat transfer coefficient, the local surface temperature, and the local film temperature as functions of x.

7–64 C&S Hot gas flows in parallel over the upper surface of a 2-m-long plate. The velocity of the gas is 17 m/s at a temperature of 250°C. The gas has a thermal conductivity of 0.03779 W/m·K, a kinematic viscosity of 3.455×10^{-5} m^2/s, and a Prandtl number of 0.6974. Two copper–silicon (ASTM B98) bolts are embedded in the plate: the first bolt at 0.5 m from the leading edge, and the second bolt at 1.5 m from the leading edge. The maximum use temperature for the ASTM B98 copper–silicon bolt is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). A cooling device removes the heat from the plate uniformly at 2000 W/m^2. Determine whether the heat being removed from the plate is sufficient to keep the bolts below the maximum use temperature of 149°C.

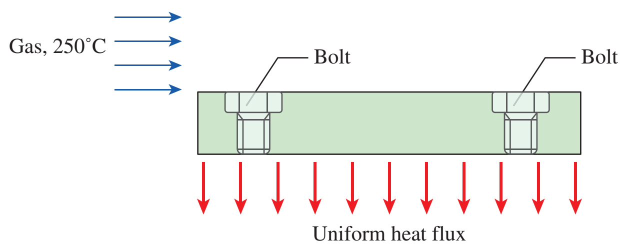

FIGURE P7–64

7–65 A 15-cm × 15-cm circuit board dissipating 20 W of power uniformly is cooled by air, which approaches the circuit board at 20°C with a velocity of 6 m/s. Disregarding any heat transfer from the back surface of the board, determine the surface temperature of the electronic components (*a*) at the leading edge and (*b*) at the end of the board. Assume the flow to be turbulent since the electronic components are expected to act as turbulators. For air properties evaluations, assume a film temperature of 35°C. Is this a good assumption?

Flow Across Cylinders and Spheres

7–66C In flow over blunt bodies such as a cylinder, how does the pressure drag differ from the friction drag?

7–67C In flow over cylinders, why does the drag coefficient suddenly drop when the flow becomes turbulent? Isn't turbulence supposed to increase the drag coefficient instead of decreasing it?

7–68C Why is flow separation in flow over cylinders delayed in turbulent flow?

7–69C Consider laminar flow of air across a hot circular cylinder. At what point on the cylinder will the heat transfer be highest? What would your answer be if the flow were turbulent?

7–70 Consider the flow of a fluid across a cylinder maintained at a constant temperature. Now the free-stream velocity of the fluid is doubled. Determine the change in the drag force on the cylinder and the rate of heat transfer between the fluid and the cylinder.

7–71 A long 12-cm-diameter steam pipe whose external surface temperature is 90°C passes through some open area that is not protected against the winds. Determine the rate of heat loss from the pipe per unit of its length when the air is at 1 atm pressure and 7°C and the wind is blowing across the pipe at a velocity of 65 km/h.

7–72 C&S Two metal plates are connected by a long ASTM B98 copper–silicon bolt. Air, at 250°C, flows at 17 m/s between the plates and across the cylindrical bolt. The diameter of the bolt is 9.5 mm, and the length of the bolt exposed to the air is 10 cm. The maximum use temperature for the ASTM B98 bolt is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). The temperature of the bolt is maintained by a cooling mechanism with the capability of removing heat at a rate of 30 W. Determine whether the heat removed from the bolt is sufficient to keep the bolt at 149°C or lower.

7–73 A heated long cylindrical rod is placed in a crossflow of air at 20°C (1 atm) with velocity of 10 m/s. The rod has a diameter of 5 mm, and its surface has an emissivity of 0.95. If the surrounding temperature is 20°C and the heat flux dissipated from the rod is 16,000 W/m^2, determine the surface temperature of the rod. Evaluate the air properties at 70°C.

7–74E A person extends his uncovered arms into the windy air outside at 54°F and 30 mph in order to feel nature closely (Fig. P7–74E). Initially, the skin temperature of the arm is 84°F. Treating the arm as a 2-ft-long and 3-in-diameter cylinder, determine the rate of heat loss from the arm.

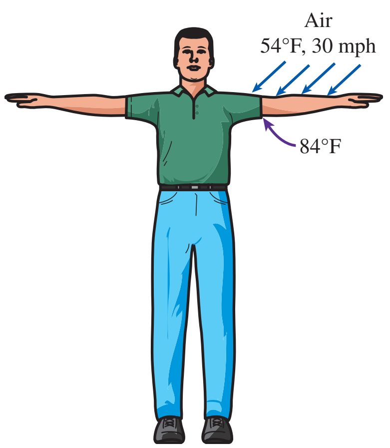

FIGURE P7–74E

7–75E Reconsider Prob. 7–74E. Using appropriate software, investigate the effects of air temperature and wind velocity on the rate of heat loss from the arm. Let the air temperature vary from 20°F to 80°F and the wind velocity from 10 mph to 40 mph. Plot the rate of heat loss as a function of air temperature and of wind velocity, and discuss the results.

7–76 In a geothermal power plant, the used geothermal water at 80°C enters a 15-cm-diameter and 400-m-long uninsulated pipe at a rate of 8.5 kg/s and leaves at 70°C before being reinjected back to the ground. Windy air at 15°C flows normal to the pipe. Disregarding radiation, determine the average wind velocity in km/h.

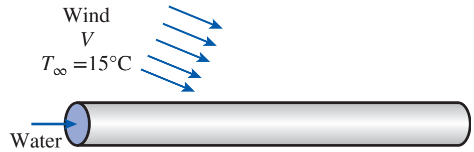

FIGURE P7–76

7–77 C&S In a piece of cryogenic equipment, two metal plates are connected by a long ASTM A437 B4B stainless steel bolt. Cold gas, at −70°C, flows between the plates and across the cylindrical bolt. The gas has a thermal conductivity of 0.02 W/m·K, a kinematic viscosity of 9.3×10^{-6} m^2/s, and a Prandtl number of 0.74. The diameter of the bolt is 9.5 mm, and the length of the bolt exposed to the gas is 10 cm. The minimum temperature suitable for the ASTM A437 B4B stainless steel bolt is −30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). The temperature of the bolt is maintained by a heating mechanism capable of providing heat at 15 W. Determine the maximum velocity that the gas can achieve without cooling the bolt below the minimum suitable temperature of −30°C.

7–78 A 5-mm-diameter electrical transmission line carries an electric current of 50 A and has a resistance of 0.002 ohm per meter length. Determine the surface temperature of the wire during a windy day when the air temperature is 10°C and the wind is blowing across the transmission line at 50 km/h.

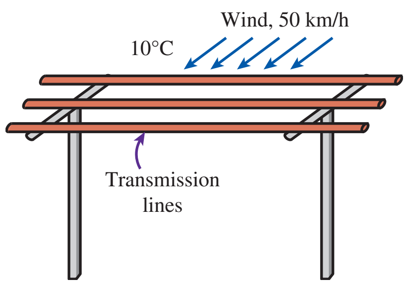

FIGURE P7–78

7–79 Reconsider Prob. 7–78. Using appropriate software, investigate the effect of the wind velocity on the surface temperature of the wire. Let the wind velocity vary from 10 km/h to 80 km/h. Plot the surface temperature as a function of wind velocity, and discuss the results.

7–80 A long aluminum wire of diameter 3 mm is extruded at a temperature of 280°C. The wire is subjected to cross airflow at 20°C at a velocity of 6 m/s. Determine the rate of heat transfer from the wire to the air per meter length when it is first exposed to the air.

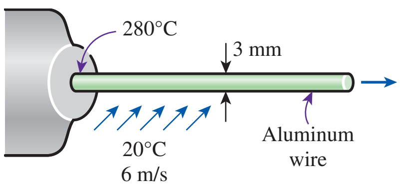

FIGURE P7–80

7–81E Consider a person who is trying to keep cool on a hot summer day by turning a fan on and exposing his entire body to airflow. The air temperature is 85°F, and the fan is blowing air at a velocity of 6 ft/s. If the person is doing light work and generating sensible heat at a rate of 300 Btu/h, determine the average temperature of the outer surface (skin or clothing) of the person. The average human body can be treated as a 1-ft-diameter cylinder with an exposed surface area of 18 ft^2. Disregard any heat transfer by radiation. What would your answer be if the air velocity were doubled? Evaluate the air properties at 100°F. *Answers:* 95.1°F, 91.6°F

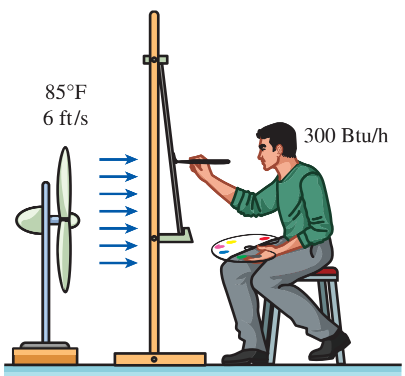

FIGURE P7–81E

7–82E A 12-ft-long, 1.5-kW electrical resistance wire is made of 0.1-in-diameter stainless steel ($k = 8.7$ Btu/h·ft·°F). The resistance wire operates in an environment at 85°F. Determine the surface temperature of the wire if it is cooled by a fan blowing air at a velocity of 20 ft/s. For evaluations of the air properties, the film temperature has to be found iteratively. As an initial guess, assume the film temperature to be 200°F.

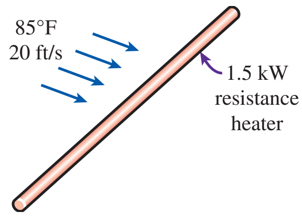

FIGURE P7–82E

7–83 A 0.4-W cylindrical electronic component with diameter 0.3 cm and length 1.8 cm and mounted on a circuit board is cooled by air flowing across it at a velocity of 240 m/min. If the air temperature is 35°C, determine the surface temperature of the component. For air properties evaluations, assume a film temperature of 50°C. Is this a good assumption?

7–84 Consider a 50-cm-diameter and 95-cm-long hot water tank. The tank is placed on the roof of a house. The water inside the tank is heated to 80°C by a flat-plate solar collector during the day. The tank is then exposed to windy air at 18°C with an average velocity of 40 km/h during the night. Estimate the temperature of the tank after a 45-min period. Assume the tank surface to be at the same temperature as the water inside and the heat transfer coefficient on the top and bottom surfaces to be the same as that on the side surface. Evaluate the air properties at 50°C.

7–85 Reconsider Prob. 7–84. Using appropriate software, plot the temperature of the tank as a function of the cooling time as the time varies from 30 min to 5 h, and discuss the results.

7–86 During a plant visit, it was noticed that a 12-m-long section of a 12-cm-diameter steam pipe is completely exposed to the ambient air. The temperature measurements indicate that the average temperature of the outer surface of the steam pipe is 75°C when the ambient temperature is 5°C. There are also light winds in the area at 25 km/h. The emissivity of the outer surface of the pipe is 0.8, and the average temperature of the surfaces surrounding the pipe, including the sky, is estimated to be 0°C. Determine the amount of heat lost from the steam during a 10-h-long workday.

Steam is supplied by a gas-fired steam generator that has an efficiency of 80 percent, and the plant pays $1.05/therm of natural gas. If the pipe is insulated and 90 percent of the heat loss is saved, determine the amount of money this facility will save a year as a result of insulating the steam pipes. Assume the plant operates every day of the year for 10 h. State your assumptions.

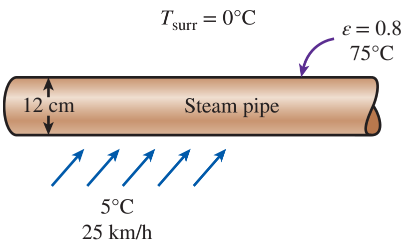

FIGURE P7–86

7–87 Reconsider Prob. 7–86. There seems to be some uncertainty about the average temperature of the surfaces surrounding the pipe used in radiation calculations, and you are asked to determine if it makes any significant difference in overall heat transfer. Repeat the calculations for average surrounding surface temperatures of −20°C and 25°C, respectively, and determine the change in the values obtained.

7–88 A 10-cm-diameter, 30-cm-high cylindrical bottle contains cold water at 3°C. The bottle is placed in windy air at 27°C. The water temperature is measured to be 11°C after 45 min of cooling. Disregarding radiation effects and heat transfer from the top and bottom surfaces, estimate the average wind velocity.

7–89 Exhaust gases from a manufacturing plant are being discharged through a 10-m-tall exhaust stack with outer diameter of 1 m. The exhaust gases are discharged at a rate of 1.2 kg/s, while temperature drop between inlet and exit of the exhaust stack is 30°C, and the constant pressure specific heat of the exhaust gases is 1600 J/kg·K. On a particular day, wind at 27°C is blowing across the exhaust stack with a velocity of 10 m/s, while the outer surface of the exhaust stack experiences radiation with the surroundings at 27°C. Solar radiation is incident on the exhaust stack outer surface at a rate of 1400 W/m^2, and both the emissivity and solar absorptivity of the outer surface are 0.9. Determine the exhaust stack outer surface temperature. Evaluate the air properties at 80°C.

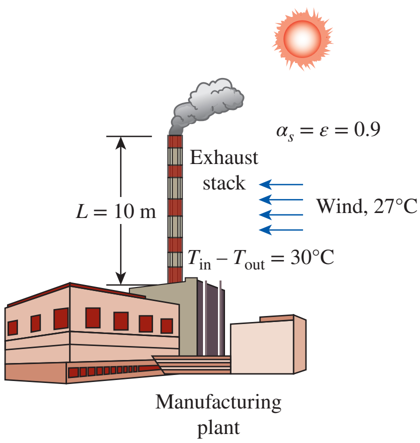

FIGURE P7–89

7–90 Exposure to high concentration of gaseous ammonia can cause lung damage. To prevent gaseous ammonia from leaking out, ammonia is transported in its liquid state through a pipe ($k = 25$ W/m·K, $D_{i,\ pipe} = 2.5$ cm, $D_{o,\ pipe} = 4$ cm, and $L = 10$ m). Since liquid ammonia has a normal boiling point of −33.3°C, the pipe needs to be properly insulated to prevent the surrounding heat from causing the ammonia to boil. The pipe is situated in a laboratory, where air at 20°C is blowing across it with a velocity of 7 m/s. The convection heat transfer coefficient of the liquid ammonia is 100 W/m^2·K. Calculate the minimum insulation thickness for the pipe using a material with k = 0.75 W/m·K to keep the liquid ammonia flowing at an average temperature of −35°C, while maintaining the insulated pipe outer surface temperature at 10°C.

7–91 Ice slurry is being transported in a pipe ($k = 15$ W/m·K, $D_i = 2.5$ cm, $D_o = 3$ cm, and $L = 5$ m) with an inner surface temperature of 0°C. The ambient condition surrounding the pipe has a temperature of 20°C and a dew point at 10°C. A blower is located near the pipe, which provides blowing air across the pipe at a velocity of 2 m/s. If the outer surface temperature of the pipe drops below the dew point, condensation can occur on the surface. Since this pipe is located near high-voltage devices, water droplets from the condensation can cause an electrical hazard. To prevent accidents, the pipe surface needs to be insulated. Calculate the minimum insulation thickness for the pipe using a material with $k = 0.95$ W/m·K to prevent the outer surface temperature from dropping below the dew point.

7–92 Air at 20°C (1 atm) is flowing over a 5-cm-diameter sphere with a velocity of 3.5 m/s. If the surface temperature of the sphere is constant at 80°C, determine (*a*) the average drag coefficient on the sphere and (*b*) the heat transfer rate from the sphere.

7–93 A stainless steel ball ($\rho = 8055$ kg/m^3, $c_p = 480$ J/kg·K) of diameter D = 15 cm is removed from the oven at a uniform temperature of 125°C. The ball is then subjected to the flow of air at 1 atm pressure and 30°C with a velocity of 6 m/s. The surface temperature of the ball eventually drops to 75°C. Determine the average convection heat transfer coefficient during this cooling process, and estimate how long this process has taken.

7–94 Reconsider Prob. 7–93. Using appropriate software, investigate the effect of air velocity on the average convection heat transfer coefficient and the cooling time. Let the air velocity vary from 1 m/s to 10 m/s. Plot the heat transfer coefficient and the cooling time as a function of air velocity, and discuss the results.

7–95 An average person generates heat at a rate of 84 W while resting. Assuming one-quarter of this heat is lost from the head and disregarding radiation, determine the average surface temperature of the head when it is not covered and is subjected to winds at 10°C and 12 km/h. The head can be approximated as a 30-cm-diameter sphere. Assume a surface temperature of 15°C for evaluation of μ_s. Is this a good assumption? *Answer:* 14.9°C

7–96 An incandescent lightbulb is an inexpensive but highly inefficient device that converts electrical energy into light. It converts about 10 percent of the electrical energy it consumes into light while converting the remaining 90 percent into heat. (A fluorescent lightbulb will give the same amount of light while consuming only one-fourth of the electrical energy, and it will last 10 times longer than an incandescent lightbulb.) The glass bulb of the lamp heats up very quickly as a result of absorbing all that heat and dissipating it to the surroundings by convection and radiation.

Consider a 10-cm-diameter, 100-W lightbulb cooled by a fan that blows air at 30°C to the bulb at a velocity of 2 m/s. The surrounding surfaces are also at 30°C, and the emissivity of the glass is 0.9. Assuming 10 percent of the energy passes through the glass bulb as light with negligible absorption and the rest of the energy is absorbed and dissipated by the bulb itself, determine the equilibrium temperature of the glass bulb. Assume a surface temperature of 100°C for evaluation of μ_s. Is this a good assumption?

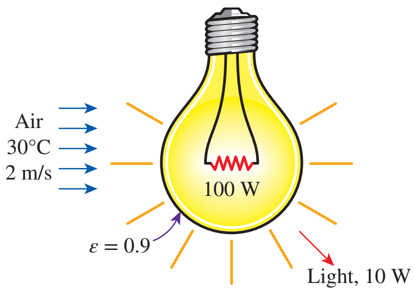

FIGURE P7–96

7–97 A 0.4-m-diameter spherical tank of negligible thickness contains iced water at 0°C. Air at 25°C flows over the tank with a velocity of 3 m/s. Determine the rate of heat transfer to the tank and the rate at which ice melts. The heat of fusion of water at 0°C is 333.7 kJ/kg.

7–98 In an experiment, the temperature of a hot airstream is to be measured by a thermocouple with a spherical junction. Due to the nature of this experiment, the response time of the thermocouple to register 99 percent of the initial temperature difference must be within 5 s. The properties of the thermocouple junction are $k = 35$ W/m·K, $\rho = 8500$ kg/m^3, and $c_p = 320$ J/kg·K. The hot air has a free-stream velocity and temperature of 3 m/s and 140°C, respectively. If the initial temperature of the thermocouple junction is 20°C, determine the thermocouple junction diameter that would satisfy the required response time of 5 s. *Hint:* Use lumped system analysis to determine the time required for the thermocouple to register 99 percent of the initial temperature difference (verify the application of this method to this problem).

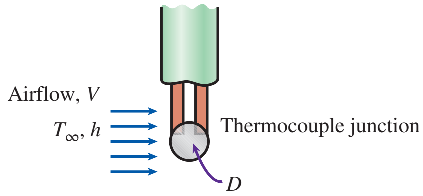

FIGURE P7–98

7–99 Reconsider Prob. 7–98. Using appropriate software, evaluate the effect of the hot air velocity on the thermocouple junction diameter that would satisfy the required response time of 5 s. By varying the hot air velocity from 1 to 10 m/s, plot the thermocouple junction diameter as a function of air velocity.

7–100 A thermocouple with a spherical junction diameter of 1 mm is used for measuring the temperature of a hydrogen gas stream. The properties of the thermocouple junction are $k = 35$ W/m·K, $\rho = 8500$ kg/m^3, and $c_p = 320$ J/kg·K. The hydrogen gas, behaving as an ideal gas at 1 atm, has a free-stream temperature of 200°C. If the initial temperature of the thermocouple junction is 10°C, evaluate the time for the thermocouple to register 99 percent of the initial temperature difference at different free-stream velocities of the hydrogen gas. Using appropriate software, perform the evaluation by varying the free-stream velocity from 1 to 100 m/s. Then, plot the thermocouple response time and the convection heat transfer coefficient as a function of free-stream velocity. *Hint:* Use the lumped system analysis to determine the time required for the thermocouple to register 99 percent of the initial temperature difference (verify the application of this method to this problem).

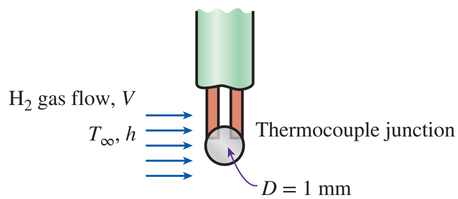

FIGURE P7–100

7–101 PtD A glass ($k = 1.1$ W/m·K) spherical tank is filled with chemicals undergoing exothermic reaction. The reaction keeps the inner surface temperature of the tank at 80°C. The tank has an inner radius of 0.5 m, and its wall thickness is 10 mm. Situated in surroundings with an ambient temperature of 15°C and a convection heat transfer coefficient of 70 W/m^2·K, the tank's outer surface is being cooled by air flowing across it at 5 m/s. In order to prevent thermal burn on individuals working around the container, it is necessary to keep the tank's outer surface temperature below 50°C. Determine whether or not the tank's outer surface temperature is safe from thermal burn hazards.

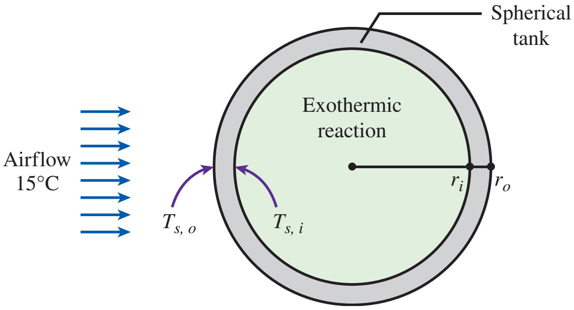

FIGURE P7–101

7–102 The components of an electronic system are located in a 1.5-m-long horizontal duct whose cross section is 20 cm × 20 cm. The components in the duct are not allowed to come into direct contact with cooling air, and thus they are cooled by air at 30°C flowing over the duct with a velocity of 200 m/min. If the surface temperature of the duct is not to exceed 65°C, determine the total power rating of the electronic devices that can be mounted into the duct.

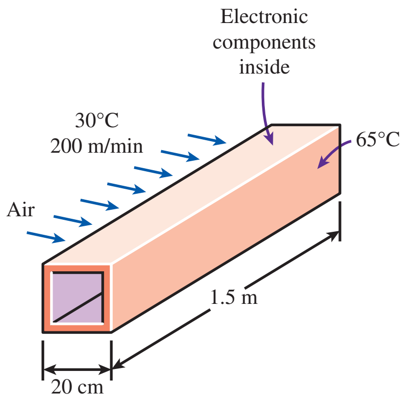

FIGURE P7–102

7–103 Repeat Prob. 7–102 for a location at a 3000-m altitude where the atmospheric pressure is 70.12 kPa.

7–104 C&S Two metal plates are connected by a long ASTM A479 904L stainless steel bar. Air, at 340°C, flows at 25 m/s between the plates and across the bar. The bar has a square cross section with a width of 10 mm, and the length of the bar exposed to the hot air is 10 cm. The maximum use temperature for the ASTM A479 904L is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The temperature of the bar is maintained by a cooling mechanism capable of removing heat at a rate of 50 W. Determine whether the heat removed from the bar is sufficient to keep the bar at 260°C or lower.

7–105 A 0.2-m × 0.2-m street sign surface has an absorptivity of 0.6 and an emissivity of 0.7, while the street sign is subjected to a crossflow wind at 20°C with a velocity of 1 m/s. Solar radiation is incident on the street sign at a rate of 1100 W/m^2, and the surrounding temperature is 20°C. Determine the surface temperature of the street sign. Evaluate the air properties at 30°C. Treat the sign surface as a vertical plate in crossflow.

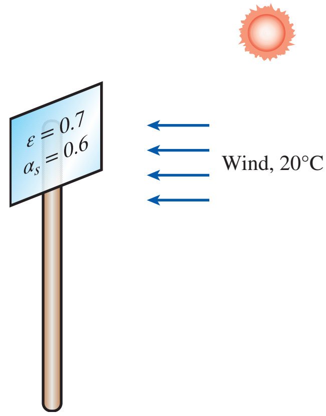

FIGURE P7–105

7–106 A coated sheet is being dried with hot air blowing in cross flow on the sheet surface. The surface temperature of the sheet is constant at 90°C, while the air velocity and temperature are 0.3 m/s and 110°C, respectively. The length of the sheet subjected to the blowing hot air is 1-m long. Determine the convection heat transfer coefficient and the heat flux added to the sheet surface. Treat the coated sheet as a vertical plate in crossflow.

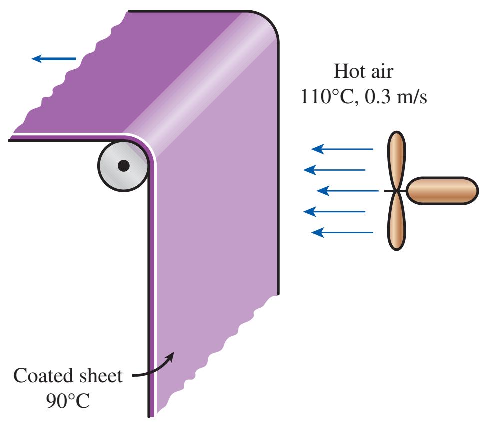

FIGURE P7–106

7–107 Reconsider Prob. 7–106. Using appropriate software, evaluate the hot air velocity on the convection heat transfer coefficient. By varying the hot air velocity from 0.15 to 0.35 m/s, plot the convection heat transfer coefficient as a function of air velocity.

Flow Across Tube Banks

7–108C In flow across tube banks, how does the heat transfer coefficient vary with the row number in the flow direction? How does it vary in the transverse direction for a given row number?

7–109C In flow across tube banks, why is the Reynolds number based on the maximum velocity instead of the uniform approach velocity?

7–110 A tube bank consists of 300 tubes at a distance of 6 cm between the centerlines of any two adjacent tubes. Air approaches the tube bank in the normal direction at 20°C and 1 atm with a mean velocity of 6 m/s. There are 20 rows in the flow direction with 15 tubes in each row with an average surface temperature of 140°C. For an outer tube diameter of 2 cm, determine the average heat transfer coefficient. Evaluate the air properties at an assumed mean temperature of 70°C and 1 atm.

7–111 Water at 15°C is to be heated to 65°C by passing it over a bundle of 4-m-long, 1-cm-diameter resistance heater rods maintained at 90°C. Water approaches the heater rod bundle in the normal direction at a mean velocity of 0.8 m/s. The rods are arranged in-line with longitudinal and transverse pitches of $S_L = 4$ cm and $S_T = 3$ cm. Determine the number of tube rows N_L in the flow direction needed to achieve the indicated temperature rise.

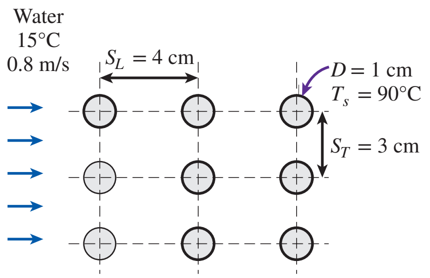

FIGURE P7–111

7–112 Air is to be heated by passing it over a bank of 3-m-long tubes inside which steam is condensing at 100°C. Air approaches the tube bank in the normal direction at 20°C and 1 atm with a mean velocity of 5.2 m/s. The outer diameter of the tubes is 1.6 cm, and the tubes are staggered with longitudinal and transverse pitches of $S_L = S_T = 4$ cm. There are 20 rows in the flow direction with 10 tubes in each row. Determine (*a*) the rate of heat transfer, (*b*) the pressure drop across the tube bank, and (*c*) the rate of condensation of steam inside the tubes. Evaluate the air properties at an assumed mean temperature of 35°C and 1 atm. Is this a good assumption?

7–113 Repeat Prob. 7–112 for in-line arrangement with $S_L = S_T = 6$ cm.

7–114 Exhaust gases at 1 atm and 300°C are used to preheat water in an industrial facility by passing them over a bank of tubes through which water is flowing at a rate of 6 kg/s. The mean tube wall temperature is 80°C. Exhaust gases approach the tube bank in the normal direction at 4.5 m/s. The outer diameter of the tubes is 2.1 cm, and the tubes are arranged in-line with longitudinal and transverse pitches of $S_L = S_T = 8$ cm. There are 16 rows in the flow direction with eight tubes in each row. Using the properties of air for exhaust gases, determine (*a*) the rate of heat transfer per unit length of tubes, (*b*) pressure drop across the tube bank, and (*c*) the temperature rise of water flowing through the tubes per unit length of tubes. Evaluate the air properties at an assumed mean temperature of 250°C and 1 atm. Is this a good assumption?

7–115 Air is to be cooled in the evaporator section of a refrigerator by passing it over a bank of 0.8-cm-outer-diameter and 0.8-m-long tubes inside which the refrigerant is evaporating at −20°C. Air approaches the tube bank in the normal direction at 0°C and 1 atm with a mean velocity of 5 m/s. The tubes are arranged in-line with longitudinal and transverse pitches of $S_L = S_T = 1.5$ cm. There are 25 rows in the flow direction with 15 tubes in each row. Determine (*a*) the refrigeration capacity of this system and (*b*) pressure drop across the tube bank. Evaluate the air properties at an assumed mean temperature of −5°C and 1 atm. Is this a good assumption?

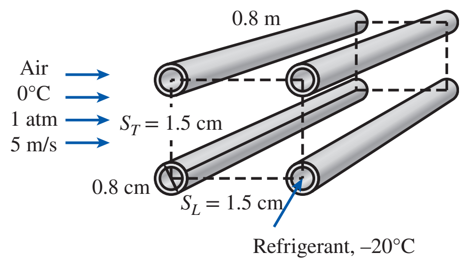

FIGURE P7–115

7–116 Repeat Prob. 7–115 by solving it for staggered arrangement with $S_L = S_T = 1.5$ cm, and compare the performance of the evaporator for the in-line and staggered arrangements.

7–117 Combustion air in a manufacturing facility is to be preheated before entering a furnace by hot water at 90°C flowing through the tubes of a tube bank located in a duct. Air enters the duct at 15°C and 1 atm with a mean velocity of 4.5 m/s, and it flows over the tubes in the normal direction. The outer diameter of the tubes is 2.2 cm, and the tubes are arranged in-line with longitudinal and transverse pitches of $S_L = S_T = 5$ cm. There are eight rows in the flow direction with eight tubes in each row. Determine the rate of heat transfer per unit length of the tubes and the pressure drop across the tube bank. Evaluate the air properties at an assumed mean temperature of 20°C and 1 atm. Is this a good assumption?

7–118 Repeat Prob. 7–117 for staggered arrangement with $S_L = S_T = 6$ cm.

Review Problems

7–119 Air at 15°C and 1 atm flows over a 0.3-m-wide plate at 65°C at a velocity of 3.0 m/s. Compute the following quantities at $x = x_{cr}$:

(*a*) Hydrodynamic boundary layer thickness, m
(*b*) Local friction coefficient
(*c*) Average friction coefficient
(*d*) Total drag force due to friction, N
(*e*) Local convection heat transfer coefficient, W/m²·K
(*f*) Average convection heat transfer coefficient, W/m²·K
(*g*) Rate of convective heat transfer, W

7–120 Air (1 atm, 5°C) with a free-stream velocity of 2 m/s flows in parallel with a stationary thin 1-m × 1-m flat plate over the top and bottom surfaces. The flat plate has a uniform surface temperature of 35°C. Determine (*a*) the average friction coefficient, (*b*) the average convection heat transfer coefficient, and (*c*) the average convection heat transfer coefficient using the modified Reynolds analogy, and compare with the result obtained in (*b*).

FIGURE P7–120

7–121 Four power transistors, each dissipating 10 W, are mounted on a thin vertical aluminum plate ($k = 237$ W/m·K) 22 cm × 22 cm in size. The heat generated by the transistors is to be dissipated by both surfaces of the plate to the surrounding air at 20°C, which is blown over the plate by a fan at a velocity of 5 m/s. The entire plate can be assumed to be nearly isothermal, and the exposed surface area of the transistor can be taken to be equal to its base area. Determine the temperature of the aluminum plate. Evaluate the air properties at a film temperature of 40°C and 1 atm.

7–122 Oil at 60°C flows at a velocity of 20 cm/s over a 5.0-m-long and 1.0-m-wide flat plate maintained at a constant temperature of 20°C. Determine the rate of heat transfer from the oil to the plate if the average oil properties are $\rho = 880$ kg/m³, $\mu = 0.005$ kg/m·s, $k = 0.15$ W/m·K, and $c_p = 2.0$ kJ/kg·K.

7–123E The passenger compartment of a minivan traveling at 70 mph can be modeled as a 3.2-ft-high, 6-ft-wide, and 12-ft-long rectangular box whose walls have an insulating value of *R*-3 (i.e., a wall thickness-to-thermal conductivity ratio of 3 h·ft²·°F/Btu). The interior of a minivan is maintained at an average temperature of 70°F during a trip at night while the outside air temperature is 90°F.

The average heat transfer coefficient on the interior surfaces of the van is 1.2 Btu/h·ft²·°F. The airflow over the exterior surfaces can be assumed to be turbulent because of the intense vibrations involved, and the heat transfer coefficient on the front and back surfaces can be taken to be equal to that on the top surface. Disregarding any heat gain or loss by radiation, determine the rate of heat transfer from the ambient air to the van. Assume the airflow to be entirely turbulent because of the intense vibrations involved. Use a film temperature of 80°F for evaluations of air properties at 1 atm.

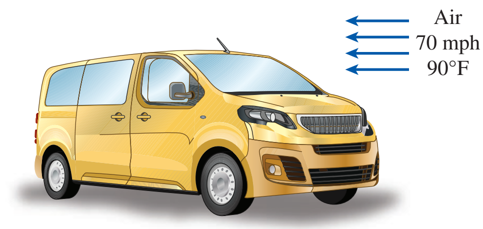

FIGURE P7–123E

7–124 A thin, square, flat plate has 1.2 m on each side. Air at 10°C flows over the top and bottom surfaces of a very rough plate in a direction parallel to one edge, with a velocity of 48 m/s. The surface of the plate is maintained at a constant temperature of 54°C. The plate is mounted on a scale that measures a drag force of 1.5 N. Determine the total heat transfer rate from the plate to the air.

7–125 Consider a house that is maintained at a constant temperature of 22°C. One of the walls of the house has three single-pane glass windows that are 1.5 m high and 1.8 m long. The glass ($k = 0.78$ W/m·K) is 0.5 cm thick, and the heat transfer coefficient on the inner surface of the glass is 8 W/m²·K. Now winds at 35 km/h start to blow parallel to the surface of this wall. If the air temperature outside is −2°C, determine the rate of heat loss through the windows of this wall. Assume radiation heat transfer to be negligible. Evaluate the air properties at a film temperature of 5°C and 1 atm.

7–126 An automotive engine can be approximated as a 0.4-m-high, 0.60-m-wide, and 0.7-m-long rectangular block. The bottom surface of the block is at a temperature of 75°C and has an emissivity of 0.92. The ambient air is at 5°C, and the road surface is at 10°C. Determine the rate of heat transfer from the bottom surface of the engine block by convection and radiation as the car travels at a velocity of 60 km/h. Assume the flow to be turbulent over the entire surface because of the constant agitation of the engine block. How will the heat transfer be affected when a 2-mm-thick layer of gunk ($k = 3$ W/m·K) has formed at the bottom surface as a result of the dirt and oil collected at that surface over time? Assume the metal temperature under the gunk is still 75°C.

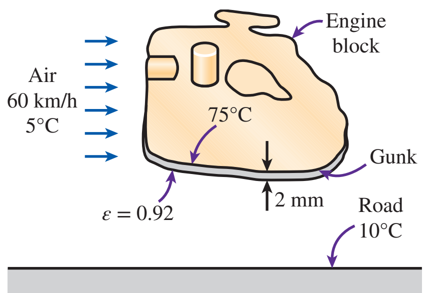

FIGURE P7–126

7–127E A 15-ft-long strip of sheet metal is being transported on a conveyor at a velocity of 16 ft/s. To cure the coating on the upper surface of the sheet metal, infrared lamps providing a constant radiant flux of 1500 Btu/h·ft^2 are used. The coating on the upper surface of the metal strip has an absorptivity of 0.6 and an emissivity of 0.7, while the surrounding ambient air temperature is 77°F. Radiation heat transfer occurs only on the upper surface, while convection heat transfer occurs on both upper and lower surfaces of the sheet metal. Determine the surface temperature of the sheet metal. Evaluate the properties of air at 180°F.

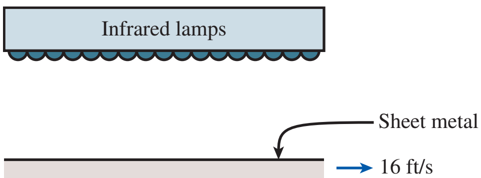

FIGURE P7–127E

7–128 To defrost ice accumulated on the outer surface of an automobile windshield, warm air is blown over the inner surface of the windshield. Consider an automobile windshield ($k_w = 1.4$ W/m·K) with an overall height of 0.5 m and thickness of 5 mm. The outside air (1 atm) ambient temperature is −20°C, and the average airflow velocity over the outer windshield surface is 80 km/h, while the ambient temperature inside the automobile is 25°C. Determine the value of the convection heat transfer coefficient for the warm air blowing over the inner surface of the windshield that is needed to cause the accumulated ice to begin melting. Assume the windshield surface can be treated as a flat plate surface.

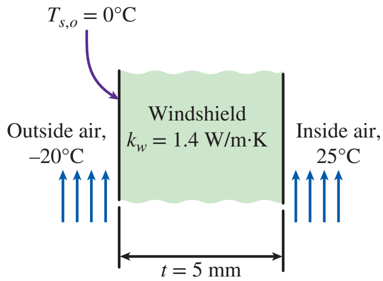

FIGURE P7–128

7–129 The local atmospheric pressure in Denver, Colorado (elevation 1610 m), is 83.4 kPa. Air at this pressure and 20°C flows with a velocity of 8 m/s over a 1.5-m × 6-m flat plate whose temperature is 140°C. Determine the rate of heat transfer from the plate if the air flows parallel to (*a*) the 6-m-long side and (*b*) the 1.5-m side.

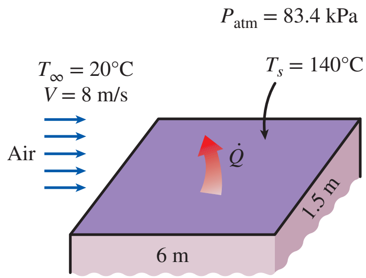

FIGURE P7–129

7–130 A 20-mm × 20-mm silicon chip is mounted such that the edges are flush in a substrate. The substrate provides an unheated starting length of 20 mm that acts as a turbulator. A turbulator is a device that trips the velocity boundary layer to turbulence. Hence the flow is turbulent over the chip. Airflow at 25°C (1 atm) with a velocity of 25 m/s is used to cool the upper surface of the chip. If the maximum surface temperature of the chip cannot exceed 75°C, determine the maximum allowable power dissipation on the chip surface.

7–131 An airstream at 1 atm flows, with a velocity of 15 m/s, in parallel over a 3-m-long flat plate where there is an unheated starting length of 1 m. The airstream has a temperature of 20°C, and the heated section of the flat plate is maintained at a constant temperature of 80°C. Determine (*a*) the local convection heat transfer coefficient at the trailing edge and (*b*) the average convection heat transfer coefficient for the heated section.

7–132 In the effort to increase the removal of heat from a hot surface at 120°C, a cylindrical pin fin ($k_f = 237$ W/m·K) with a diameter of 5 mm is attached to the hot surface. Air at 20°C (1 atm) is flowing across the pin fin with a velocity of 10 m/s.

Determine the maximum possible rate of heat transfer from the pin fin. Evaluate the air properties at 70°C.

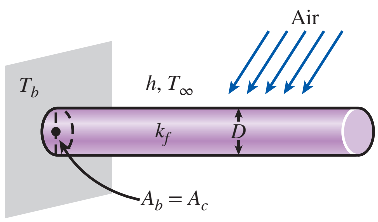

FIGURE P7–132

7–133 Consider a person who is trying to keep cool on a hot summer day by turning a fan on and exposing his body to airflow. The air temperature is 32°C, and the fan is blowing air at a velocity of 5 m/s. The surrounding surfaces are at 40°C, and the emissivity of the person can be taken to be 0.9. If the person is doing light work and generating sensible heat at a rate of 90 W, determine the average temperature of the outer surface (skin or clothing) of the person. The average human body can be treated as a 30-cm-diameter cylinder with an exposed surface area of 1.7 m^2. Evaluate the air properties at a film temperature of 35°C and 1 atm. *Answer:* 36.2°C

7–134E A transistor with a height of 0.32 in and a diameter of 0.22 in is mounted on a circuit board. The transistor is cooled by air flowing over it at a velocity of 600 ft/min. If the air temperature is 120°F and the transistor case temperature is not to exceed 180°F, determine the amount of power this transistor can dissipate safely.

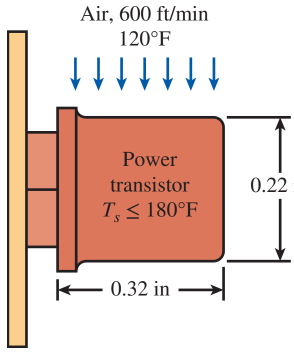

FIGURE P7–134E

7–135 Steam at 250°C flows in a stainless steel pipe ($k = 15$ W/m·K) whose inner and outer diameters are 4 cm and 4.6 cm, respectively. The pipe is covered with 3.5-cm-thick glass wool insulation ($k = 0.038$ W/m·K) whose outer surface has an emissivity of 0.3. Heat is lost to the surrounding air and surfaces at 3°C by convection and radiation. Taking the heat transfer coefficient inside the pipe to be 80 W/m^2·K, determine the rate of heat loss from the steam per unit length of the pipe when air is flowing across the pipe at 4 m/s. Evaluate the air properties at a film temperature of 10°C and 1 atm.

7–136 A cylindrical rod is placed in a crossflow of air at 20°C (1 atm) with velocity of 10 m/s. The rod has a diameter of 5 mm and a constant surface temperature of 120°C. Determine (*a*) the average drag coefficient, (*b*) the convection heat transfer coefficient using the Churchill and Bernstein relation, and (*c*) the convection heat transfer coefficient using Table 7–1.

7–137 A 0.55-m-internal-diameter spherical tank made of 1-cm-thick stainless steel ($k = 15$ W/m·K) is used to store iced water at 0°C. The tank is located outdoors at 30°C and is subjected to winds at 8 km/h. Assuming the entire steel tank to be at 0°C and thus its thermal resistance to be negligible, determine (*a*) the rate of heat transfer to the iced water in the tank and (*b*) the amount of ice at 0°C that melts during a 24-h period. The heat of fusion of water at atmospheric pressure is $h_{if} = 333.7$ kJ/kg. Disregard any heat transfer by radiation.

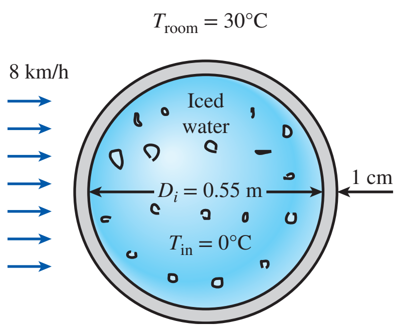

FIGURE P7–137

7–138 Repeat Prob. 7–137, assuming the inner surface of the tank to be at 0°C but by taking the thermal resistance of the tank and heat transfer by radiation into consideration. Assume the average surrounding surface temperature for radiation exchange to be 25°C and the outer surface of the tank to have an emissivity of 0.75. *Answers:* (*a*) 379 W, (*b*) 98.1 kg

7–139 An array of electrical heating elements is used in an air-duct heater as shown in Fig. P7–139. Each element has a length of 200 mm and a uniform surface temperature of 350°C. Atmospheric air enters the heater with a velocity of 8 m/s and a temperature of 25°C. Determine the total heat transfer rate and the temperature of the air leaving the heater. Neglect the change in air properties as a result of temperature change across the heater. Evaluate the air properties at an assumed mean temperature of 35°C and 1 atm. Is this a good assumption?

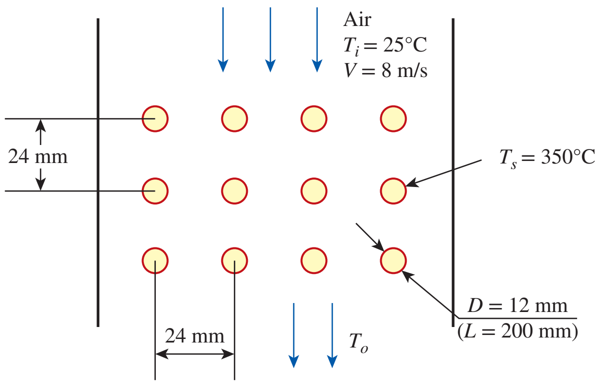

FIGURE P7–139

Fundamentals of Engineering (FE) Exam Problems

7–140 For laminar flow of a fluid along a flat plate, one would expect the largest local convection heat transfer coefficient for the same Reynolds and Prandl numbers when

(*a*) The same temperature is maintained on the surface
(*b*) The same heat flux is maintained on the surface
(*c*) The plate has an unheated section
(*d*) The plate surface is polished
(*e*) None of the above

7–141 Air at 20°C flows over a 4-m-long and 3-m-wide surface of a plate whose temperature is 80°C with a velocity of 7 m/s. The length of the surface for which the flow remains laminar is

(*a*) 0.9 m (*b*) 1.3 m (*c*) 1.8 m
(*d*) 2.2 m (*e*) 3.7 m

(For air, use $k = 0.02735$ W/m·K, Pr = 0.7228, $\nu = 1.798 \times 10^{-5}$ m^2/s)

7–142 Air at 20°C flows over a 4-m-long and 3-m-wide surface of a plate whose temperature is 80°C with a velocity of 5 m/s. The rate of heat transfer from the laminar flow region of the surface is

(*a*) 950 W (*b*) 1037 W (*c*) 2074 W
(*d*) 2640 W (*e*) 3075 W

(For air, use $k = 0.02735$ W/m·K, Pr = 0.7228, $\nu = 1.798 \times 10^{-5}$ m^2/s)

7–143 Engine oil at 105°C flows over the surface of a flat plate whose temperature is 15°C with a velocity of 1.5 m/s. The local drag force per unit surface area 0.8 m from the leading edge of the plate is

(*a*) 21.8 N/m^2 (*b*) 14.3 N/m^2 (*c*) 10.9 N/m^2
(*d*) 8.5 N/m^2 (*e*) 5.5 N/m^2

(For oil, use $\nu = 8.565 \times 10^{-5}$ m^2/s, $\rho = 864$ kg/m^3)

7–144 Air ($k = 0.028$ W/m·K, Pr = 0.7) at 50°C flows along a 1-m-long flat plate whose temperature is maintained at 20°C with a velocity such that the Reynolds number at the end of the plate is 10,000. The heat transfer per unit width between the plate and air is

(*a*) 20 W/m (*b*) 30 W/m (*c*) 40 W/m
(*d*) 50 W/m (*e*) 60 W/m

7–145 Air at 15°C flows over a flat plate subjected to a uniform heat flux of 240 W/m^2 with a velocity of 3.5 m/s. The surface temperature of the plate 6 m from the leading edge is

(*a*) 40.5°C (*b*) 41.5°C (*c*) 58.2°C
(*d*) 95.4°C (*e*) 134°C

(For air, use $k = 0.02551$ W/m·K, Pr = 0.7296, $\nu = 1.562 \times 10^{-5}$ m^2/s)

7–146 Air at 20°C flows over a 4-m-long and 3-m-wide surface of a plate whose temperature is 80°C with a velocity of 5 m/s. The rate of heat transfer from the surface is

(*a*) 7383 W (*b*) 8985 W (*c*) 11,231 W
(*d*) 14,672 W (*e*) 20,402 W

(For air, use $k = 0.02735$ W/m·K, Pr = 0.7228, $\nu = 1.798 \times 10^{-5}$ m^2/s)

7–147 Water at 75°C flows over a 2-m-long, 2-m-wide surface of a plate whose temperature is 5°C with a velocity of 1.5 m/s. The total drag force acting on the plate is

(*a*) 2.8 N (*b*) 12.3 N (*c*) 13.7 N
(*d*) 15.4 N (*e*) 20.0 N

(For water, use $\nu = 0.658 \times 10^{-6}$ m^2/s, $\rho = 992$ kg/m^3)

7–148 Air at 25°C flows over a 5-cm-diameter, 1.7-m-long smooth pipe with a velocity of 4 m/s. A refrigerant at −15°C flows inside the pipe, and the surface temperature of the pipe is essentially the same as the refrigerant temperature inside. The drag force exerted on the pipe by the air is

(*a*) 0.4 N (*b*) 1.1 N (*c*) 8.5 N
(*d*) 13 N (*e*) 18 N

(For air, use $\nu = 1.382 \times 10^{-5}$ m^2/s, $\rho = 1.269$ kg/m^3)

7–149 Air at 25°C flows over a 4-cm-diameter, 1.7-m-long pipe with a velocity of 4 m/s. A refrigerant at −15°C flows inside the pipe, and the surface temperature of the pipe is essentially the same as the refrigerant temperature inside. Air properties at the average temperature are $k = 0.0240$ W/m·K, Pr = 0.735, $\nu = 1.382 \times 10^{-5}$ m^2/s. The rate of heat transfer to the pipe is

(*a*) 126 W (*b*) 245 W (*c*) 302 W
(*d*) 415 W (*e*) 556 W

7–150 Kitchen water at 10°C flows over a 10-cm-diameter pipe with a velocity of 1.1 m/s. Geothermal water enters the pipe at 90°C at a rate of 1.25 kg/s. For calculation purposes, the surface temperature of the pipe may be assumed to be 70°C. If the geothermal water is to leave the pipe at 50°C, the required length of the pipe is

(*a*) 1.1 m (*b*) 1.8 m (*c*) 2.9 m
(*d*) 4.3 m (*e*) 7.6 m

(For both water streams, use $k = 0.631$ W/m·K, Pr = 4.32, $\nu = 0.658 \times 10^{-6}$ m²/s, $c_p = 4179$ J/kg·K)

7–151 Wind at 30°C flows over a 0.5-m-diameter spherical tank containing iced water at 0°C with a velocity of 25 km/h. If the tank is thin-shelled with a high thermal conductivity material, the rate at which ice melts is

(*a*) 4.78 kg/h (*b*) 6.15 kg/h (*c*) 7.45 kg/h
(*d*) 11.8 kg/h (*e*) 16.0 kg/h

(Take $h_{if} = 333.7$ kJ/kg, and use the following for air: $k = 0.02588$ W/m·K, Pr = 0.7282, $\nu = 1.608 \times 10^{-5}$ m²/s, $\mu_\infty = 1.872 \times 10^{-5}$ kg/m·s, $\mu_s = 1.729 \times 10^{-5}$ kg/m·s)

7–152 Ambient air at 20°C flows over a 30-cm-diameter hot spherical object with a velocity of 4.2 m/s. If the average surface temperature of the object is 200°C, the average convection heat transfer coefficient during this process is

(*a*) 8.6 W/m·K (*b*) 15.7 W/m·K (*c*) 18.6 W/m·K
(*d*) 21.0 W/m·K (*e*) 32.4 W/m·K

(For air, use $k = 0.2514$ W/m·K, Pr = 0.7309, $\nu = 1.516 \times 10^{-5}$ m²/s, $\mu_\infty = 1.825 \times 10^{-5}$ kg/m·s, $\mu_s = 2.577 \times 10^{-5}$ kg/m·s)

7–153 Jakob (1949) suggests the following correlation be used for square tubes in a liquid crossflow situation:

$$\text{Nu} = 0.102\ \text{Re}^{0.675}\ \text{Pr}^{1/3}$$

Water ($k = 0.61$ W/m·K, Pr = 6) at 50°C flows across a 1-cm-square tube with a Reynolds number of 10,000 and surface temperature of 75°C. If the tube is 3 m long, the rate of heat transfer between the tube and water is

(*a*) 9.8 kW (*b*) 12.4 kW (*c*) 17.0 kW
(*d*) 19.6 kW (*e*) 24.0 kW

7–154 Air (Pr = 0.7, $k = 0.026$ W/m·K) at 200°C flows across 3-cm-diameter tubes whose surface temperature is 50°C with a Reynolds number of 8000. The Churchill and Bernstein (1977) convective heat transfer correlation for the average Nusselt number in this situation is

$$\text{Nu} = 0.3 + \frac{0.62\ \text{Re}^{0.5}\ \text{Pr}^{0.33}}{[1 + (0.4/\text{Pr})^{0.67}]^{0.25}}$$

(*a*) 1.3 kW/m² (*b*) 2.4 kW/m² (*c*) 4.1 kW/m²
(*d*) 5.7 kW/m² (*e*) 8.2 kW/m²

7–155 Jakob (1949) suggests the following correlation be used for square tubes in a liquid crossflow situation:

$$\text{Nu} = 0.102\ \text{Re}^{0.625}\ \text{Pr}^{1/3}$$

Water ($k = 0.61$ W/m·K, Pr = 6) flows across a 1-cm-square tube with a Reynolds number of 10,000. The convection heat transfer coefficient is

(*a*) 5.7 kW/m²·K (*b*) 8.3 kW/m²·K (*c*) 11.2 kW/m²·K
(*d*) 15.6 kW/m²·K (*e*) 18.1 kW/m²·K

Design and Essay Problems

7–156 On average, superinsulated homes use just 15 percent of the fuel required to heat the same size conventional home built before the energy crisis in the 1970s. Write an essay on superinsulated homes, and identify the features that make them so energy efficient as well as the problems associated with them. Do you think superinsulated homes will be economically attractive in your area?

7–157 Conduct this experiment to determine the heat loss coefficient of your house or apartment in W/°C or Btu/h·°F. First make sure that the conditions in the house are steady and the house is at the set temperature of the thermostat. Use an outdoor thermometer to monitor outdoor temperature. One evening, using a watch or timer, determine how long the heater was on during a 3-h period and the average outdoor temperature during that period. Then using the heat output rating of your heater, determine the amount of heat supplied. Also, estimate the amount of heat generation in the house during that period by noting the number of people, the total wattage of lights that were on, and the heat generated by the appliances and equipment. Using that information, calculate the average rate of heat loss from the house and the heat loss coefficient.

7–158 The decision whether to invest in an energy-saving measure is made on the basis of the length of time needed for it to pay for itself in projected energy (and thus cost) savings. The easiest way to decide is to calculate the simple payback period by simply dividing the installed cost of the measure by the annual cost savings and comparing it to the lifetime of the installation. This approach is adequate for short payback periods (less than five years) in stable economies with low interest rates (under 10 percent) since the error involved is no larger than the uncertainties. However, if the payback period is long, it may be necessary to consider the *interest rate* if the money is to be borrowed, or the *rate of return* if the money is invested elsewhere instead of the energy conservation measure. For example, a simple payback period of five years corresponds to 5.0, 6.12, 6.64, 7.27, 8.09, 9.919, 10.84, and 13.91 for an interest rate (or return on investment) of 0, 6, 8, 10, 12, 14, 16, and 18 percent, respectively. Finding out the proper relations from engineering economics books, determine the payback periods for the interest rates given above corresponding to simple payback periods of 1 to 10 years.